Sticky Hierarchies: How Caste Norms Bound Gender Progress in India*

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Abstract

The broad brush explanation for all gender gaps is "patriarchal social norms", but when multiple social hierarchies coexist, is it possible that patriarchy is being defined by another hierarchy, which constrains women's outcomes? We study this question in the context of India and show that gender norms improved substantially over two decades (1998-2021), but unevenly across regions and caste groups. We proxy caste norms with Hindu dietary practices and instrument contemporary diet with the 1931 Brahman and Jain population shares in an overidentified LIML specification. Within the caste-purity belt of northern, western, central, and southern India, an interquartile weakening of caste dietary orthodoxy narrows the spousal education gap by about 1.9 years, raises female education by about 1.6 years, and reduces son preference. The effect on female labour force participation is statistically indistinguishable from zero, consistent with demand-side explanations for its decline. Our results suggest that gender equality and the dismantling of caste hierarchy are complements, not separable agendas.

Keywords: compound hierarchies, caste, gender norms, intersectionality, female labour force participation, social change, India

JEL Codes: J16, J15, O15, Z13, I24

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1 Introduction

In many of the world's largest developing economies, including India, Turkey, Egypt, and Indonesia, female labour force participation (FLFP) has fallen or stagnated during decades of rapid economic growth and measurably improving gender attitudes, and progress on other gender outcomes has been uneven across regions and social groups. The standard toolkit of gender economics, which attributes women's disadvantage to patriarchal social norms, offers no satisfying explanation: if norms are improving, why are outcomes worsening? Thus, the bulk of the literature assumes patriarchal norms in these countries to be either sticky or deteriorating.

A large literature in sociology and feminist theory argues that gender cannot be analysed in isolation from race, class, caste, and ethnicity: hierarchies compound (see, e.g., Crenshaw, 1989; Collins, 2000). Yet the economics literature on gender in developing countries has been slow to take this insight to the data. Studies of gender gaps routinely *control* for caste or ethnicity as confounders, but they rarely ask whether caste norms, the beliefs and practices that enforce social hierarchy, independently shape the constraints women face. We provide the first systematic evidence that they do.

We argue that the uneven pace of gender progress reflects the fact that gender norms do not evolve independently of caste norms. They are shaped and bounded by the caste norms that govern purity, pollution, endogamy, and social standing. Where caste orthodoxy is strong, the gender norms that govern marriage, education, and women's agency are correspondingly constrained; where caste orthodoxy has weakened, gender norms have room to move. Caste is, in this sense, the *sticky* hierarchy. Attempts to shift gender norms without also shifting the caste hierarchy that bounds them may therefore under-deliver, a concern that applies wherever gender intersects with another axis of stratification.

India is a useful setting for these questions. It combines large sub-national variation in both caste and gender norms with repeated, nationally representative surveys covering hundreds of thousands of women. Crucially, our findings are not merely descriptive of India: they speak directly to debates about compound hierarchies in other stratified societies, including the race-gender intersectionality literature in the United States (Crenshaw, 1989), scholarship on ethnic hierarchies and women's outcomes in sub-Saharan Africa (Jayachandran, 2015) and class-gender interactions in Latin America.

This paper makes three contributions. First, we document that gender norms in India have improved substantially and heterogeneously. The most prominent recent descriptive analysis of South Asia, by Bussolo et al. (2022), concludes that there has been almost no progress on gender equality in the region over the past half-century, with education the

lone exception, and that gender attitudes have remained unchanged or in places become more conservative. Using individual-level data from Rounds 2 (1998–99) and 5 (2019–21) of the National Family Health Survey, covering over 630,000 ever-married women (the population both rounds interviewed), and disaggregating by region and caste, we reach a different conclusion for India. Between the two rounds, average female education rose by more than three years, from 3.5 to 6.7, the ages at first marriage and first birth each rose by almost two years, stated son preference (the ideal number of boys minus girls) fell by more than half, from 0.40 to 0.16, stated justification of domestic violence fell by 12 percentage points, and women's reported say over their own healthcare and mobility expanded by 27 and 51 percentage points respectively. Norms improve on most margins in every region and caste group, but the *pace* of improvement differs: the North–South divide runs in different directions for different norms, and caste gradients are large and persistent. The one margin on which we concur with the earlier literature is female labour force participation, which does not improve.

Second, and centrally, we present evidence that caste norms shape gender outcomes. We proxy caste norm strength primarily with non-vegetarian dietary practices among Hindus, drawn from the Household Consumption Expenditure Survey, and complement it with the prevalence of untouchability practices (India Human Development Survey data) that are illegal, but continue covertly as well as overtly; and the incidence of hate crimes against formerly untouchable castes, the Scheduled Castes (using official data from the National Crime Records Bureau). Districts with weaker dietary orthodoxy have narrower spousal education gaps, lower son preference, and greater mobility agency. Crimes against Scheduled Castes, measured at the district level, give the same pattern across education, marriage age, son preference, the spousal education gap, and agency; the state-level untouchability measure points in the same direction but is imprecise. To move from association to causation, we instrument contemporary dietary practice with two historically determined features of local caste demography from the pre-Independence colonial census: the 1931 Brahman population share (Castelló-Climent et al., 2017) and the 1931 Jain population share (Fenske et al., 2025). Brahmans and Jains are both historically vegetarian, but their geographies of settlement are distinct, which allows us to combine them in an overidentified limited-information maximum likelihood (LIML) specification that is less biased than 2SLS when instruments are weak and that admits a test of the overidentifying restriction. The joint first-stage F-statistic is 19.6 for our preferred fish-and-meat measure, 24.8 for upper-caste consumption, and 8.7 for the narrower fish-only measure, and the Hansen J test does not reject for any gender outcome ($p \geq 0.15$). The design identifies the effect of persistent, *cross-sectional* differences in caste orthodoxy across districts, the variation that crystallised before 1931, rather than short-run change within a district. Throughout, these are local effects, identified within the caste-purity

belt where the diet proxy measures caste orthodoxy.

The most stable result concerns the marriage market. An interquartile weakening of caste dietary orthodoxy narrows the spousal education gap by 1.85 years and raises average female education by 1.57 years. On the sample of couples, the gap closes because wives' education rises; husbands' education does not change detectably. The result is consistent with the relaxation of caste endogamy, which restricts a woman's marriage pool to her own caste, but our data cannot separate that channel from a direct effect of caste norms on women's schooling. Weaker caste norms also reduce son preference and widen the spousal age gap, because men's age at marriage rises by more than women's. We interpret the latter as consistent with a shift from kin-arranged marriage, in which both spouses marry at customary ages, to marriage that rests on the groom's own earning capacity, which accrues with age. The coefficients on stated justification and on experienced domestic violence both run the other way; they may reflect more candid reporting where caste norms are weaker or a real increase in violence, and we treat them as an open question.

Third, our results bear on the FLFP puzzle. Female labour force participation in India—as in Turkey, Egypt, and several other middle-income countries—fell during a period of rapid growth and improving gender attitudes, before partially recovering. Some attribute this to a cultural backlash or to the failure of norm change to translate into labour market outcomes (Jensen, 2012; Klasen, 2019). Companion work (Deshpande and Singh, 2021, 2024) suggests instead that the trajectory reflects structural transformation on the demand side. Our evidence is consistent with that reading: attitudinal indicators improve throughout the period of declining participation, and we do not detect a causal effect of caste norms on FLFP. Because participation is an equilibrium of labour supply and demand, we do not read this null as proof that caste norms leave women's labour supply untouched; it is consistent with caste norms operating upstream, on education and marriage, while labour demand determines whether women's supply is absorbed into measured work.

Taken together, the findings suggest that single-axis explanations for women's disadvantage that focus on gender norms alone may misdiagnose the problem when a second hierarchy is operative, and that gender equality and the weakening of hierarchical social structures are complements.

The remainder of the paper proceeds as follows. Section 2 sets out our conceptual framework and relates the paper to the literature. Section 3 describes the data. Section 4 documents how gender norms changed across regions and caste groups, and how they co-evolved with caste norms. Section 5 presents cross-sectional evidence on caste norms and gender outcomes. Section 6 sets out the identification strategy, and Section 7 reports the causal estimates, mechanisms, robustness, and heterogeneity. Section 8 turns to female

labour force participation. Section 9 concludes.

2 Conceptual Framework and Related Literature

The answer to "what are gender norms?" depends on one's theoretical starting point. A large strand of the economics literature treats norms as coordination equilibria: rules that individuals comply with because they expect others to comply, and expect others to expect compliance in turn. This individualist, expectations-based account has tractability advantages (it generates predictions about tipping points and information interventions), but it has serious limitations for the purposes of this paper.

The main limitation is that the expectations account is depoliticized. It treats norms as equilibria without asking whose interests they serve, what ideological superstructure sustains them, or which institutions enforce them. It focuses analysis on the attitudes and behaviours of ordinary people, drawing attention away from the structural inequalities embedded in institutions. It also conflates categorically different types of norm: a norm about which side of the road to drive on and a norm about which gender inherits land or which caste may enter a temple are not the same kind of social fact. The former is a pure coordination device; the latter is an ideological rule that allocates power and privilege across a social hierarchy. As UN Women (2026) notes, social norms on gender "operate very differently to those on table manners, smoking or handwashing": they are rules about the distribution of power as well as expectations.

We follow instead a framework that defines social norms as embedded rules, standards, and practices that structure relationships between people and social institutions (the state, market, community, family, and religion), based on intersecting ideologies that uphold social hierarchies, including patriarchy and casteism (UN Women, 2026; Bedford and Brosio, 2024). Three elements of this framework are essential for this paper.

(a) **Ideologies.** The norms implicated in gender inequality are ideological rules rather than mere conventions. Patriarchy, the ideological claim to the supremacy and naturalness of cisgendered male authority, prescribes where women are placed in social institutions and what power and privileges they are entitled to. In the Indian context, this ideology does not operate in isolation: it intersects with casteism, the organizing ideology of caste hierarchy rooted in concepts of purity and pollution. Caste ideology does more than add to patriarchal constraints: it defines the space within which gender norms can evolve. Where caste orthodoxy is strong, sustained by strict dietary prohibitions, untouchability practices, and the enforcement of endogamy, the marriage market, labour market, and educational opportunities available to women are structurally bounded by the caste hierarchy, and gender norms must operate within those bounds. This intersection between compound ideological hierarchies is

the central empirical claim of our paper.

(b) **Institutions.** Norms become 'social' through the institutions that organize society (UN Women, 2026; reproduced in Appendix Figure A2). These institutions, namely the state, market, community, family, and religion, do not merely reflect norms; they produce, reproduce, and enforce them, and they form nexuses of mutual reinforcement. A girl's exclusion from education is not simply a 'family norm' but is embedded in the rules and practices of the wider community, and religious authority acting in concert. Similarly, the dietary practices that index caste orthodoxy in our analysis are not merely private household choices; they are publicly visible signals of caste adherence, enforced through community surveillance and social sanction. This institutional character is what makes caste norms 'sticky' in the sense we argue: they are not attitudinal positions that individuals hold but rules embedded in multiple interlocking institutions, which is why nearly a century of legal and policy reform, from the abolition of untouchability to affirmative action (Bertrand et al., 2010) and constitutional guarantees, has not dislodged them in large swaths of India. Persistence here reflects institutional embeddedness rather than inertia.

(c) **Power.** The production, reproduction, and transformation of social norms is an exercise in power (UN Women, 2026; reproduced in Appendix Figure A2). Those who hold power in social institutions, in the Indian case upper-caste men who disproportionately control the institutions of state, religion, and community, define prevailing norms and deploy their institutional position to enforce them. Those at the bottom of intersecting hierarchies of gender and caste have the fewest resources for resistance, and bear the highest costs of deviation. This power dimension has two implications for our analysis. First, the persistence of caste norms across nine decades, captured by our historical instruments, reflects the institutional dominance of groups that benefit from caste hierarchy, not mere cultural inertia. Second, the improvements in gender outcomes we document over two decades represent a contested terrain in which countervailing forces have partially succeeded in shifting norms against institutional resistance.

2.1 Measurement

The UN Women framework proposes four sets of indicators for gauging whether norm change has occurred (UN Women, 2026; reproduced in Appendix Figure A1): shifts in narratives across institutional sites; changes in the material conditions of women's lives; the strength of countervailing power and collective action; and gender equality outcomes. Our data speak most directly to the fourth set. The NFHS indicators we analyze, including female education, marriage age, son preference, domestic violence,

and agency, are gender equality outcome indicators: they reveal whether conditions in women's lives have shifted in ways consistent with norm change. They are not direct measures of narratives or institutional rules.

This is both a limitation and a strength of our approach. A limitation, because individual-level survey measures capture only surface-level responses, are sensitive to social desirability pressures, and systematically under count the institutional dimensions of norm change (Bedford and Brosio, 2024). We are alert to this throughout: our treatment of the domestic violence justification coefficient in the IV specification (Section 7.5) is a case where the survey measure must be read carefully, because social desirability pressures may vary with caste orthodoxy: women in more orthodox districts may be less likely to voluntarily report acceptance of stigmatized attitudes, regardless of underlying beliefs. A strength, because gender equality outcomes, the actual conditions of women's lives across five institutional sites, are the ultimate criterion by which norm change must be judged. Narrative shifts and the growth of countervailing power matter insofar as they eventually register in outcomes; our data on over 630,000 ever-married women across two decades provide the evidentiary basis for whether they have.

The dietary proxy for caste norms also merits interpretation through this lens. Dietary practices are not private preferences that happen to be observable; they are publicly visible acts of conformity to or deviation from caste hierarchy, enforced through the institutions of community and family. Their distribution across districts captures the continued grip of caste ideology on these institutional sites. The historical instruments (the 1931 Brahman and Jain population shares) capture the depth of this ideological-institutional entrenchment: settlement patterns crystallized before Indian independence, before reservations, before the legal prohibition of untouchability, and they still predict dietary practices and gender outcomes nine decades later. That persistence is the empirical content of our claim that caste is the sticky hierarchy.

2.2 Related Literature

Two recent papers are closest to ours. Agte and Bernhardt (2024) show that caste purity norms function as an investment in social status: where Hindus live alongside Adivasis, an indigenous group outside the caste system, adherence to caste rules weakens and Hindu women become substantially more likely to work. Munshi and Singh (2024) offer a status-signalling account of the reverse movement, in which households withdraw women from the labour force to signal wealth, so that female labour force *non-participation* rises as the economy grows. Both papers are about caste, status, and women's *work*: the labour-supply margin is the outcome of interest, and the mechanism runs through the status value of female seclusion. Our paper differs on three counts. First, our object of study is the full set of gender norms rather than labour supply alone, and the outcomes

on which caste norms have the clearest effects are pre-labour-market margins (female education, the marriage market, and son preference) that shape women's opportunities before any work decision is taken. Second, we do not detect a causal effect of caste norms on female labour force participation itself. Third, our identification is different in kind: rather than out-group contact at the village level or within-district change over time, we exploit pre-Independence caste demography to recover the effect of persistent, historically rooted caste orthodoxy.

3 Data

3.1 Gender Norms

Our primary data source is the National Family Health Survey (NFHS), a large-scale nationally representative household survey conducted by the Ministry of Health and Family Welfare (IIPS and ORC Macro, 2000; IIPS and ICF, 2021).[1] Five rounds have been fielded since 1992–93; we compare Round 2 (1998–99) and Round 5 (2019–21), a span of roughly two decades that brackets a period of rapid economic and social change. NFHS-2 interviewed only ever-married women aged 15–49, whereas NFHS-5 interviewed all women aged 15–49, so every comparison across the two rounds restricts NFHS-5 to ever-married women. The harmonised sample contains 633,133 ever-married women: 90,303 in NFHS-2 and 542,830 in NFHS-5.

From the women's questionnaire we construct the following gender norm indicators:

- *Education and reproductive timing:* years of education; age at first marriage; age at first birth; spousal education gap; spousal age gap.
- *Son preference:* the difference between the respondent's ideal number of boys and ideal number of girls.
- *Domestic violence:* whether the respondent has ever been physically hurt by her husband or in-laws; whether she believes a husband is justified in beating his wife in at least one of the three scenarios asked in both rounds (she goes out without telling him, neglects the house or children, or does not cook properly).
- *Female agency:* whether the respondent has the final say in decisions about her own healthcare; whether she can move about freely outside the home (mobility agency).

NFHS is not a labor force survey and is administered to women between 15 and 49 years; thus, the labor force participation estimates are not comparable to official estimates. Therefore, we constructed female labor force participation estimates at the district level from the Employment-Unemployment Survey (EUS) 2011-12, which we mapped to gender

[1] NFHS was initially part of the Demographic and Health Surveys supported by USAID. It has subsequently been conducted by the Ministry of Health and Family Welfare, Government of India.

norm variables from NFHS-4 (2015-16), and the Periodic Labor Force Survey (PLFS) 2019, which is mapped to gender norms from NFHS-5 (2019-21).

Eligibility also differs by question (Appendix Table A6). NFHS-2 asked every respondent all of these questions; NFHS-5 asked husband's characteristics, decision-making and DV justification only in its state-module subsample, and experienced violence only in a private domestic violence module. We define each indicator on its eligible population, restrict the domestic violence measures to questions common to both rounds, and weight all estimates. Table 1 reports summary statistics.

Table 1: Summary statistics for select gender norms, ever-married women, NFHS-2 and NFHS-5

	NFHS-2 (1998–99)					NFHS-5 (2019–21)				
	Mean	SD	P25	P75	N	Mean	SD	P25	P75	N
Women's education (years)	3.51	4.54	0	7	90,265	6.72	5.31	0	10	542,830
Age at first marriage	16.85	3.15	15	18	90,303	18.63	3.86	16	21	542,187
Age at first birth	18.75	3.41	16	21	80,869	20.58	3.79	18	23	493,135
Son preference (ideal boys − girls)	0.40	0.74	0	1	85,501	0.16	0.55	0	0	536,796
DV experienced	0.19	0.40	0	0	90,294	0.28	0.45	0	1	63,851
DV justified	0.48	0.50	0	1	89,759	0.36	0.48	0	1	81,211

Notes: Ever-married women aged 15–49 (NFHS-2 interviewed only ever-married women). Statistics are weighted; N is the unweighted number of respondents. DV experienced: ever physically hurt by husband or in-laws (NFHS-5: DV-module respondents, DV weight). DV justified: beating justified in at least one of three scenarios asked in both rounds. Eligibility by indicator is in Appendix Table A6.

3.2 Caste Norms

India's caste system consists of several thousand *jatis* or castes. The formerly untouchable castes and several marginalized tribal communities are the most socioeconomically disadvantaged groups in the country. These groups receive preferential affirmative action, for the purpose of which these are listed in a government schedule. Thus, this collection of castes and tribes are referred to as the Scheduled Castes and Scheduled Tribes (SC and ST). While these are official, administrative categories, SCs often self-identify themselves as *Dalit*, meaning "oppressed", as a term of pride. *Adivasi* means "indigenous people" and is often used to describe tribal communities. A group of intermediate to low-ranked castes and communities, which also receive affirmative action, are called the "Other Backward Classes" (OBCs). All those not classified as SC, ST or OBC are the residual "Others" category. The caste system can be seen in all major religions in India, but we restrict our sample to Hindu individuals, as Hinduism has the most formalised caste system. Thus, in our case, the "Others" category captures Hindu Upper Castes (non-SC-ST-OBC Hindus).

We use district-level diet measures as a proxy for caste norms in our analysis. Upper-caste Hindu traditions strongly proscribe non-vegetarian food, and the strength of these prohibitions varies across regions in ways that align with the historical depth of caste hierarchy.

Two caveats about the proxy matter for everything that follows. First, the dietary proxy indexes caste orthodoxy only where the caste hierarchy is organised around Brahmanical vegetarian purity. In the eastern fish-eating belt and in Kashmir, upper-caste status does not map onto vegetarianism, so a low non-vegetarian share there reflects regional cuisine rather than caste orthodoxy; Section 6 explains how our estimation sample excludes these regions. Second, the proxy captures caste orthodoxy as a whole, indexed by diet rather than exhausted by it.

For constructing the measures, we use the National Sample Survey Office's Household Consumption Expenditure Survey (HCES) across three rounds: HCES 1999–2000 (NSS 55th round), HCES 2011-12 (NSS 68th round, Schedule 1.0 Type-2), and HCES 2023-24. The causal analysis uses the 2011–12 and 2023–24 rounds.

Each observation is a household-item record reporting consumption of a specific food item over a 30-day reference period. The relevant non-vegetarian item codes included are 191–196[2]:

- 191: Fish and prawn
- 192–196: Goat/mutton, beef/buffalo, pork, chicken, other meat

Our three proxies are then:

1. `%eat(fish)`: the share of Hindu households consuming fish (item code 191) or the residual "other" non-vegetarian category (item code 196), whether or not they also consume meat.
2. `%eat(fish+meat)`: the share of Hindu households consuming any non-vegetarian item (codes 191–196).
3. `%UC_eat(fish+meat)`: the same as the second measure but restricted to upper-caste Hindu households, identified as those in the "Others" social group in the NSS classification (non-SC/ST/OBC).

The proxy ranges from 0 (all Hindu households purely vegetarian) to 1 (all consume meat/fish). Lower values indicate stronger caste-based dietary norms in the district. We consider the three proxies to be monotonically increasing in restrictiveness, with the upper-caste only measure (identified as those with "Others" social group in NSS, i.e., non-SC/ST/OBC) being the most stringent diet measure.

Each proxy is constructed the same way from the household-item records. We first restrict to Hindu households, and, for the upper-caste measure, further restrict to households in the "Others" social group. For each household we then form an indicator equal to one if it

[2]Eggs (item code 190) are deliberately excluded because they may not be considered non-vegetarian.

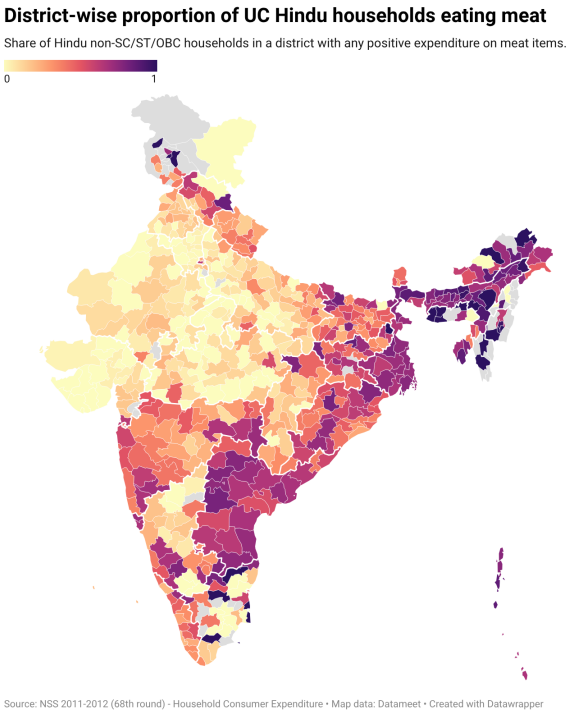


Figure 1: District-wise proportion of upper-caste Hindu households consuming meat (NSS HCES 2011–12). Districts in yellow have near-zero non-vegetarian consumption among upper-caste Hindus, indicating strong caste dietary orthodoxy. Districts in dark purple have high consumption shares, indicating weaker orthodoxy. The North–South gradient is consistent with the regional embeddedness of caste norms as institutional rules rather than individual preferences.

reports consuming any of the relevant item codes over the 30-day reference period, taking the maximum of the item-level consumption indicators within the household. Finally, we average this household indicator across all households in a district to obtain the district-level share. The procedure is carried out separately for each HCES round.

The causal analysis pairs district-level gender outcomes from NFHS-4 (2015–16) with caste norms measured in 2011–12, and we replicate it with NFHS-5 (2019–21) and caste norms measured in 2023–24. The co-evolution analysis in Section 4.4 uses three paired rounds: NFHS-2 with HCES 1999–2000, NFHS-4 with HCES 2011–12, and NFHS-5 with HCES 2023–24.

Figure 1 maps the proxy across India's districts. Caste dietary orthodoxy, as captured by near-universal vegetarianism among upper-caste Hindus, is concentrated in the Indo-Gangetic North and Northwest, while the South, East, and Northeast show substantially weaker dietary restrictions. This regional gradient is consistent with the embeddedness of Brahmanical vegetarianism in the institutions of community and family in the regions where upper-caste political and religious authority has historically been strongest.

We complement the diet measure with two proxies that do not depend on food taboos. The first is the prevalence of untouchability practices, drawn from the India Human Development Survey (IHDS-2, 2011) and available at the state level. The second is the

incidence of registered crimes against Scheduled Castes per 10,000 population, drawn from National Crime Records Bureau data for 2019. These proxies address the concern that dietary patterns capture regional cuisine rather than caste hierarchy. We use diet as the main proxy because it is measured consistently at the district level across rounds and because it admits the historical instruments developed below.

3.3 Controls

Controls are drawn from the SHRUG (Socioeconomic High-resolution Rural-Urban Geographic Platform; Asher et al., 2021), which provides population figures (from Census 2011) and night-light density at the district-level.

Agricultural Employment Share is constructed by summing cultivators and agricultural labourers (both main and marginal workers) and dividing by total workers. This captures the structural composition of district-level employment.

The Scheduled Tribe population share is the ST population divided by total district population, both taken from the SHRUG's 2011 Primary Census Abstract. It enters every cross-sectional and IV specification as a control, because ST populations sit outside the caste hierarchy, follow distinct dietary and gender practices, and are unevenly distributed across the districts where our instruments vary.

The SHRUG's Primary Census Abstract does not provide population figures divided by type of region (urban/rural) at the district-level. To calculate urban share of population for each district, we thus go back to the Primary Census Abstract of the 2011 Census, which reports total/urban/rural population for every district.

SHRUG's annual Visible Infrared Imaging Radiometer Suite (VIIRS) night-light dataset provides a continuous and standardized proxy for district-level economic activity. We restrict to year 2012 (the closest VIIRS year to our HCES 2011-12 caste-norm proxy) and the median-masked category, which removes top-coded values (industrial flares, gas burns) that would otherwise dominate the district mean. The variable used is `viirs_annual_mean`, the per-cell mean luminosity within the district. We add 1 before taking the log to handle zero-light districts.

4 Gender Norms over Time: Regional and Caste Variation

We organise the evidence around two broad clusters: (i) education, marriage, and reproductive timing norms; and (ii) son preference, domestic violence, and female agency norms.

A note on the sample is in order, because it differs by design from the causal analysis later in the paper. This descriptive section uses the *full national sample*, all states and

all caste groups, and does so deliberately. Within it, we restrict to ever-married women aged 15–49, the population both NFHS rounds interviewed, and weight all estimates (Section 3). The improvement in gender norms over the two decades is not confined to particular regions or communities: on most margins it occurs across every region and every caste group, even as its *pace* varies. Documenting that nationwide improvement is the point of this section, and restricting the sample would obscure it. The later analysis of *caste norms* restricts to the mechanism-valid caste-purity belt, where our diet proxy and historical instruments are meaningful; the two samples answer two different questions, and we keep them distinct.

4.1 Regional Variation across Time

To assess how gender norms evolved differently across India's regions, we estimate:

$$y_{irt} = \beta_0 + \beta_1 \,\text{SW-Region}_{ir} + \beta_2 \,\text{Round}_t + \beta_3 \,(\text{SW-Region} \times \text{Round})_{irt} + \mathbf{X}_i' + \varepsilon_{irt} \quad (4.1)$$

where y_{irt} is the gender norm for individual i in region r and survey round t; SW-Region is a dummy for India's South-West (Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Goa, Kerala, Tamil Nadu, Telangana, and the union territories of Daman & Diu, Dadra & Nagar Haveli, Puducherry, Lakshadweep, and Andaman & Nicobar Islands); Round equals one for NFHS-5; and $\mathbf{X}_i$ are individual controls (age, own education, spouse's education, and paid-work status) included for the attitudinal outcomes. The coefficient β_3 captures the differential change in the South-West relative to the North.

Table 2: Women's education, marriage and birth age norms across region and time

	(1) Female education (years)	(2) Age at marriage	(3) Age at first birth	(4) Spousal education gap	(5) Spousal age gap
South/West States	-0.66	-0.55	-0.21	1.52***	-1.42***
	(0.61)	(0.41)	(0.32)	(0.30)	(0.29)
Round 5	2.50***	1.45***	1.66***	-0.83**	-1.74***
	(0.45)	(0.38)	(0.33)	(0.39)	(0.52)
South/West States × Round 5	2.32***	1.22**	0.59	-2.48***	2.54***
	(0.74)	(0.57)	(0.50)	(0.46)	(0.66)
Observations	633095	632490	574004	171363	161461

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). Spousal gaps use currently married women; in NFHS-5, husband's education and age come from the state-module subsample. All variables are reported in years. SE reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The headline findings are as follows.

In Northern states, ever-married women in NFHS-5 had 2.50 more years of education, married 1.45 years later, and had their first child 1.66 years later than in NFHS-2 (all statistically significant at 1%). Spousal education and age gaps both narrowed, by 0.83 and 1.74 years. There are no statistically significant differences in female education, marriage age, or first-birth age between South-West and Northern states at baseline. South-West states improved faster: female education rose by a further 2.32 years ($p <$

0.01), age at marriage by a further 1.22 years ($p < 0.05$), and the spousal education gap narrowed by a further 2.48 years ($p < 0.01$), though the spousal age gap diverged ($\hat{\beta}_3 = +2.54$, $p < 0.01$).

Table 3: Son preference, domestic violence & women's agency norms across region and time

	(1) Son preference	(2) DV experienced	(3) DV justified	(4) Health agency	(5) Mobility agency
South/West States	0.30***	0.03	-0.05	-0.04	-0.10**
	(0.06)	(0.02)	(0.06)	(0.03)	(0.04)
Round 5	-0.09	0.13***	-0.18***	0.27***	0.51***
	(0.06)	(0.03)	(0.06)	(0.03)	(0.03)
South/West States × Round 5	-0.40***	-0.05	0.33***	-0.03	0.05
	(0.07)	(0.05)	(0.11)	(0.04)	(0.04)
Age	0.00**	0.00	-0.00***	0.01***	0.01***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Female education (years)	-0.02***	-0.01***	-0.01***	0.01***	0.01***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Spouse education (years)	-0.00	-0.01***	-0.01***	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Paid work	-0.05***	0.08***	0.06***	0.06***	0.07***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Observations	165736	153653	170431	166718	166705

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). DV experienced is the share ever physically hurt by the husband or in-laws, estimated on DV-module respondents with the DV weight (d005); sexual violence is excluded because NFHS-2 did not ask about it. DV justified uses the three scenarios asked in both rounds. In NFHS-5, the control for husband's education and the agency and DV-justification questions come from the state-module subsample. Son preference is the difference between reported ideal number of boys and girls. All other outcomes are binary indicators of reported experience or attitudes. SE reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In Northern states, DV justification fell by 18 percentage points and female agency in health and mobility increased by 27 and 51 percentage points respectively (all statistically significant at 1%), while stated son preference (the difference between the ideal numbers of boys and girls) fell by a statistically non-significant 0.09. South-West states had higher baseline son preference (+0.30, $p < 0.01$) but reduced it by a further 0.40 by 2019–21 ($p < 0.01$). DV justification diverges across regions: it fell in Northern states, while in the South-West the differential ($\hat{\beta}_3 = +0.33$, $p < 0.01$) more than offsets the Northern decline. Reported experience of violence by husbands or in-laws rose by 13 percentage points in Northern states ($p < 0.01$; nationally from 0.19 to 0.28, Table 1). Because NFHS-5 asks these questions in a separate module administered in private, the rise may partly reflect greater willingness or opportunity to report rather than a true rise in incidence; the data cannot separate the two.

4.2 Variation across Caste Groups over Time

The disaggregations in this subsection and the next are the first, descriptive look at the paper's central claim. If caste structures the constraints women face, then gender outcomes and their trajectories should differ systematically across caste groups, and those caste differences should themselves vary across regions whose caste histories differ. The

caste-group analysis here, and the region-by-caste triple difference that follows, establish that these patterns are present in the raw data before we turn to the question of whether the relationship is causal in Section 7. They motivate, rather than test, the causal argument: large and persistent caste gaps that move little over two decades are exactly what a sticky-caste-hierarchy account predicts.

Next, we disaggregate by caste group, estimating:

$$y_{ist} = \beta_0 + \beta_1 \,\text{Caste}_i + \beta_2 \,\text{Round}_t + \beta_3 \,(\text{Caste} \times \text{Round})_{it} + \alpha_s + \mathbf{X}_i' + \varepsilon_{ist} \tag{4.2}$$

where Caste is a categorical variable with Scheduled Caste (SC), Scheduled Tribe (ST), and Other Backward Class (OBC) as categories and Upper caste as the reference group; α_s are state fixed effects. The coefficient β_3 captures the differential change over time for each caste group relative to Upper caste.

Table 4: Women's education, marriage and birth age norms, across caste groups and time

	(1) Female education (years)	(2) Age at marriage	(3) Age at first birth	(4) Spousal education gap	(5) Spousal age gap
SC	-2.67***	-1.14***	-1.01***	-0.06	0.06
	(0.26)	(0.17)	(0.13)	(0.19)	(0.07)
ST	-3.42***	-1.13***	-1.08***	-0.37*	0.10
	(0.26)	(0.20)	(0.23)	(0.21)	(0.17)
OBC	-1.64***	-0.63**	-0.48*	0.18	-0.06
	(0.40)	(0.31)	(0.24)	(0.17)	(0.15)
Round 5	3.69***	1.92***	2.02***	-1.30***	-1.57***
	(0.50)	(0.41)	(0.35)	(0.35)	(0.56)
SC × Round 5	-0.10	0.06	0.01	0.40*	-0.35*
	(0.45)	(0.18)	(0.15)	(0.20)	(0.19)
ST × Round 5	-0.40	0.11	0.11	0.80***	-0.81***
	(0.42)	(0.27)	(0.30)	(0.26)	(0.28)
OBC × Round 5	-0.13	-0.12	-0.19	0.14	-0.15
	(0.56)	(0.35)	(0.29)	(0.19)	(0.26)
Observations	603010	602449	546726	166425	156776

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). Spousal gaps use currently married women; in NFHS-5, husband's education and age come from the state-module subsample. All variables are reported in years. SE reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results in Tables 4 and 5 reveal a clear caste gradient in baseline outcomes: SC women have 2.67 fewer years of education, marry 1.14 years earlier, and have their first child 1.01 years earlier than Upper caste women (all $p < 0.01$). ST and OBC women face similar disadvantages of varying magnitudes. Despite these large baseline gaps, the rate of improvement over time is broadly similar across groups—the gaps are not closing, but neither are they widening for most outcomes. One exception is the spousal education gap, which converged more slowly for SC women ($\hat{\beta}_3 = +0.40$ years, $p < 0.1$) and ST women ($+0.80$, $p < 0.01$) than for Upper caste women, while the spousal age gap narrowed faster for ST women (-0.81, $p < 0.01$). For DV justification, ST and OBC women show greater convergence, each with a differential fall of 7 percentage points ($p < 0.05$).

Table 5: Son preference, female agency & domestic violence norms across caste groups and time

	(1) Son preference	(2) DV experienced	(3) DV justified	(4) Health agency	(5) Mobility agency
SC	0.06***	0.04***	0.04**	-0.02	0.02**
	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)
ST	0.02	0.01	0.05**	-0.00	0.04*
	(0.02)	(0.01)	(0.02)	(0.03)	(0.02)
OBC	0.01	0.00	0.06***	-0.02*	0.02
	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)
Round 5	-0.13*	0.08*	-0.09*	0.24***	0.55***
	(0.07)	(0.04)	(0.05)	(0.03)	(0.04)
SC × Round 5	-0.05	0.01	-0.03	0.03	-0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
ST × Round 5	-0.01	0.02	-0.07**	0.04	-0.03
	(0.04)	(0.02)	(0.04)	(0.04)	(0.04)
OBC × Round 5	0.01	0.02	-0.07**	0.03*	-0.02
	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	160922	149575	165527	162003	161991

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). DV experienced is the share ever physically hurt by the husband or in-laws, estimated on DV-module respondents with the DV weight (d005); sexual violence is excluded because NFHS-2 did not ask about it. DV justified uses the three scenarios asked in both rounds. In NFHS-5, the control for husband's education and the agency and DV-justification questions come from the state-module subsample. Son preference is the difference between reported ideal number of boys and girls. All other outcomes are binary indicators of reported experience or attitudes. Controls: woman's age, own education (years), spouse's education (years) and paid-work status. All columns include state fixed effects. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.3 Triple Difference: Region, Caste, and Time

To examine whether caste-group trajectories differ by region, we estimate a triple-difference specification:

$$\begin{aligned} y_{irt} = {} & \beta_0 + \beta_1 \,\text{Caste}_i + \beta_2 \,\text{Round}_t + \beta_3 \,\text{Region}_r \\ & + \beta_4 \,(\text{Caste} \times \text{Round}) + \beta_5 \,(\text{Caste} \times \text{Region}) + \beta_6 \,(\text{Round} \times \text{Region}) \\ & + \beta_7 \,(\text{Caste} \times \text{Region} \times \text{Round}) + \varepsilon_{irt} \end{aligned} \tag{4.3}$$

estimated without individual controls on ever-married women in NFHS-2 and NFHS-5, weighted, with standard errors clustered by state. The triple-difference coefficient β_7 identifies whether a particular caste group in the South-West had a differential trend over time compared with the same group in the North. Appendix Tables A7 and A8 report the full estimates.

Selected statistically significant results are:

- OBC and SC women in the South-West reduced their spousal education gap by 1.84 and 1.24 years more than their Northern counterparts (relative to Upper caste; both $p < 0.01$), pointing to more rapid norm convergence in the South-West.
- OBC women in the South-West also gained 2.37 more years of education and married 1.58 years later than OBC women in the North (both $p < 0.05$), though their spousal age gap widened by 1.26 years ($p < 0.05$).
- Son preference fell faster among SC and OBC women in the South-West, by 0.14 and 0.20 more than in the North (both $p < 0.05$).

4.4 Co-evolution of Caste and Gender Norms

Before turning to cross-sectional and causal estimates, it is useful to see how caste norms and gender norms have moved together over the period we study. We restrict throughout this subsection to the mechanism-valid sample of states used in our instrumental-variables analysis, dropping Bihar, West Bengal, Jammu & Kashmir, and the eight Northeast states, where the diet proxy does not track caste orthodoxy; the co-evolution we document is therefore for exactly the population on which our causal estimates are identified. Figure 2 indexes the national means of both sets of measures to 100 in the NFHS-2/HCES 1999–2000 baseline and tracks them across the three rounds for which we can pair a diet round with a norms round. The broader diet proxies drift upward, that is, caste dietary orthodoxy weakens over two decades, and the gender outcomes move in the directions that signal progress over the same window, with the agency and education measures rising and son preference, DV justification, and the spousal gaps falling. Caste norms and gender norms loosen in tandem. The fish-only measure is flat because the main fish-eating states

are excluded from this sample and the coastal states are already near saturation; the rise in the broader measure, from 0.29 to 0.45 of Hindu households, comes from meat.

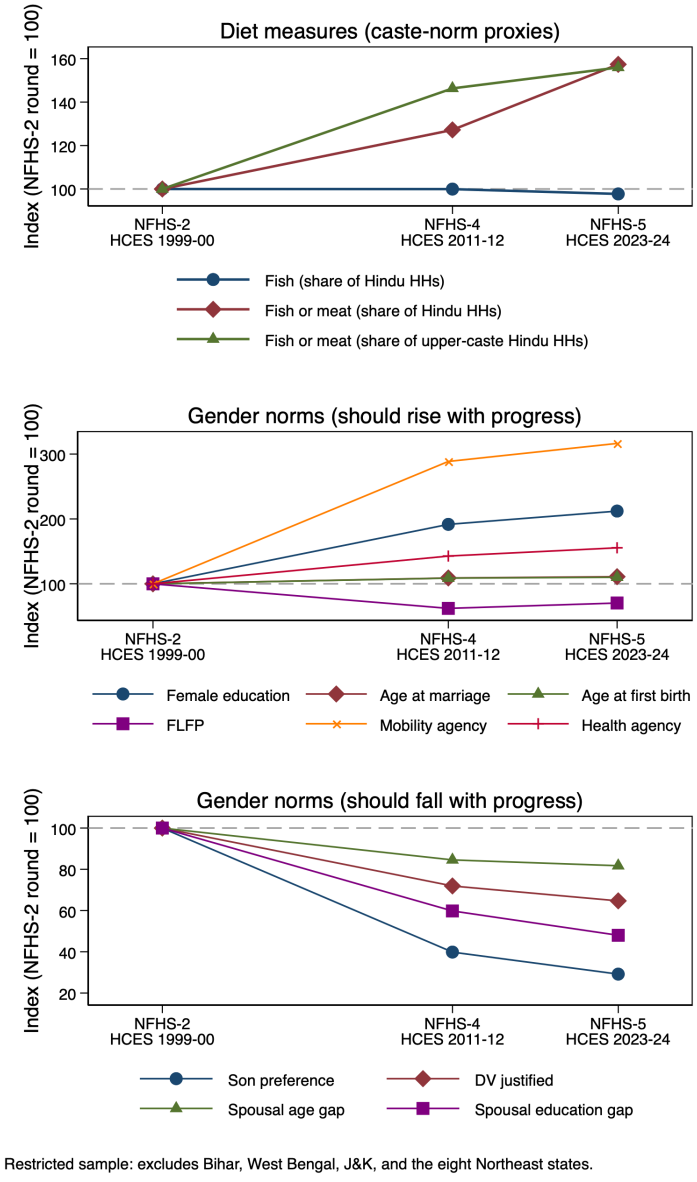


Figure 2: Co-evolution of caste norms (diet) and gender norms over time, national-mean indices with the NFHS-2/HCES 1999–2000 baseline set to 100, restricted to the mechanism-valid sample (excludes Bihar, West Bengal, Jammu & Kashmir, and the eight Northeast states). Top panel: the three diet-based caste-norm proxies. Middle panel: gender outcomes that should rise with progress (education, marriage and first-birth age, FLFP, mobility and health agency). Bottom panel: gender outcomes that should fall with progress (son preference, DV justification, spousal age and education gaps).

Figure 3 makes the same point as a trajectory, on the same restricted sample. Each panel plots the national mean of one gender outcome against the share of Hindu households consuming fish or meat (our preferred caste-norm proxy), with the three rounds shown as separate points. As caste orthodoxy weakens (movement rightward along the horizontal axis), female education and marriage age rise and son preference, the spousal age gap, FLFP non-participation, and DV justification fall. The district-level analogue of this

relationship, estimated round by round, is shown in Appendix Figure A3; it suggests that the co-movement is not a purely national artefact but is present in the cross-section of districts within each round.

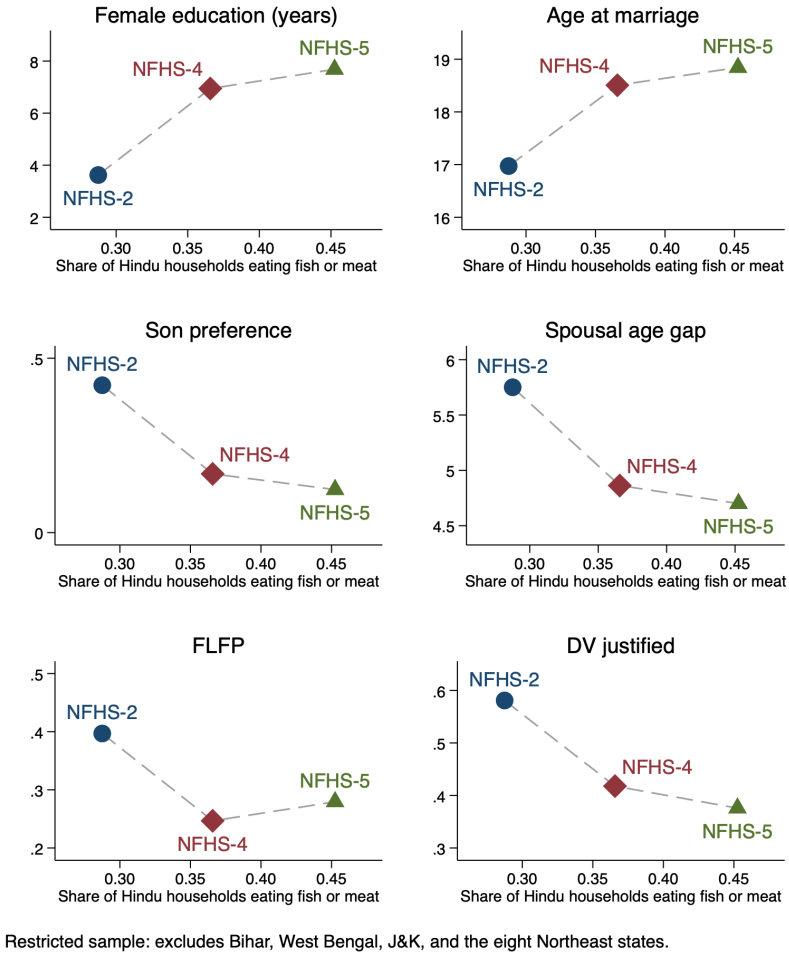


Figure 3: National co-evolution of diet (share of Hindu households eating fish or meat) and each gender outcome, restricted to the mechanism-valid sample (excludes Bihar, West Bengal, Jammu & Kashmir, and the eight Northeast states). Each panel is one gender norm; the three points trace the NFHS-2/HCES 1999–2000, NFHS-4/HCES 2011–12, and NFHS-5/HCES 2023–24 rounds. The dashed line is the fitted trajectory.

These descriptive trajectories do not establish causation: caste and gender norms could be co-moving in response to common forces such as urbanisation or rising incomes. Disentangling that is the task of the identification strategy below. They do, however, motivate the central question of the paper, namely whether the weakening of caste norms causally drives the improvement in gender outcomes that accompanies it.

5 Caste Norms and Gender Outcomes: Cross-Sectional Evidence

The central proposition of this paper is that caste norms—the hierarchical social rules governing purity, pollution, inter-group interaction, and social standing associated with

the caste system—are a distinct and empirically important determinant of gender outcomes, over and above the patriarchal norms that have received the bulk of attention in the literature. This section documents the cross-sectional association between caste norms and gender outcomes before we turn to identification.

5.1 Diet-Based Caste Norms and Gender Outcomes

Table 6: Cross-sectional effects of caste norms on gender norms, excluding Bihar, West Bengal, and Jammu and Kashmir

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish										
% eat (fish)	0.268	0.462	0.174	-0.124	0.226***	-0.0199	2.202***	-1.469***	0.0368	0.0215
	(0.587)	(0.470)	(0.385)	(0.0817)	(0.0721)	(0.0357)	(0.484)	(0.421)	(0.0381)	(0.0347)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	541	541	541	519	541	541	541	541	541	541
R-squared	0.481	0.495	0.427	0.176	0.156	0.204	0.231	0.371	0.210	0.132
Panel B: Share of Hindu HHs eating fish or meat										
% eat (fish+meat)	0.609	0.255	-0.275	0.0255	0.324***	-0.0903**	3.163***	-1.955***	0.0559*	0.0188
	(0.573)	(0.435)	(0.309)	(0.0873)	(0.0794)	(0.0377)	(0.315)	(0.419)	(0.0306)	(0.0289)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	541	541	541	519	541	541	541	541	541	541
R-squared	0.487	0.490	0.431	0.158	0.274	0.243	0.437	0.473	0.223	0.132
Panel C: Share of Upper-caste Hindu HHs eating fish or meat										
% UC eat (fish+meat)	0.450	0.203	-0.201	0.00371	0.251***	-0.0700**	2.711***	-1.542***	0.0475*	0.0206
	(0.460)	(0.379)	(0.288)	(0.0699)	(0.0638)	(0.0334)	(0.270)	(0.340)	(0.0272)	(0.0250)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	527	527	527	505	527	527	527	527	527	527
R-squared	0.500	0.495	0.435	0.153	0.219	0.236	0.393	0.424	0.188	0.104

Notes: District-level cross-section. Gender norms from NFHS-4 (2015-16), caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). Controls: district 2011 population (Census 2011), agricultural employment share, urban share, scheduled tribe population share, and log light density 2012. Sample excludes Bihar, West Bengal, and Jammu and Kashmir (Brahmin meat consumption in these states biases the diet proxy of caste norms). Standard errors clustered at the state level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6 reports our central cross-sectional result: district-level correlations between each caste-norm diet measure and gender outcomes, conditional on district population, agricultural employment share, urban share, the Scheduled Tribe population share, and log night-light density. A natural worry about such relationships is that they are driven by the handful of states where the diet proxy does not track caste orthodoxy, namely the eastern fish-eating belt and Kashmir, where upper-caste Hindus conventionally eat fish or meat. The table therefore drops the three states for which this concern is greatest: Bihar, West Bengal, and Jammu & Kashmir. For the fish-and-meat proxy (Panel B), weaker caste norms are associated with a narrower spousal education gap (-1.96 years, $p < 0.01$), a wider spousal age gap ($+3.16$ years, $p < 0.01$), greater mobility agency ($+0.056$, $p < 0.10$), lower son preference (-0.090, $p < 0.05$), and higher *stated* justification of domestic violence ($+0.32$, $p < 0.01$), the same reporting pattern we return to in the IV estimates below. The association with health agency ($+0.019$) is positive but not statistically significant. With the Scheduled Tribe share included as a control, the spousal-gap, son-preference and DV-justification gradients remain large and statistically significant, while the agency gradients are smaller and less precisely estimated. That the cross-sectional gradients survive dropping the mechanism-invalid states is the descriptive

counterpart to the identification-region argument in Section 6: the diet–gender relationship is a feature of the caste-purity belt rather than an artefact of pooling regions where the proxy means different things.

The full cross-sectional grid, which pools all states and both survey rounds (HCES 2011–12 paired with NFHS-4 and HCES 2023–24 paired with NFHS-5), is reported in Appendix Table A5. With round dummies only, a one-unit increase in the district share of Hindu households eating fish and meat is associated with 0.56 additional years of female education, 0.91 additional years of marriage age, 0.08 higher mobility agency, a 1.90-year narrower spousal education gap, and 0.07 lower son preference (all statistically significant at 1%). Adding district fixed effects, so that only within-district change between rounds identifies the coefficient, shrinks most of these estimates toward zero; Section 7.6 discusses this contrast.

5.2 Determinants of Caste Norms

Before instrumenting for caste norms, it helps to understand what determines their strength across districts, both because the determinants are of interest in their own right and because they motivate the choice of instrument. We examine two predictors: caste-group fractionalization and the Scheduled Tribe population share.

Table 7: Correlation of non-veg diet (caste norms) with caste-group fractionalization at district-level (2011-12)

	(1) % eat (fish)	(2) % eat (fish+meat)	(3) % UC eat (fish+meat)
Caste fractionalization index	-0.17* (0.09)	-0.35*** (0.09)	-0.33*** (0.10)
Observations	585	585	578

Notes: Sample: Hindu households, district-level. Source: EUS 2011-12 (caste fractionalization index, one minus the sum of squared shares of ST, SC, OBC and other Hindu households; higher values are more fractionalized); HCES 2011-12 (diet proxy). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Correlation of non-veg diet (2023-24) with caste-group fractionalization at district-level (2011-12)

	(1) % eat (fish)	(2) % eat (fish+meat)	(3) % UC eat (fish+meat)
Caste fractionalization index	-0.38*** (0.10)	-0.54*** (0.10)	-0.68*** (0.12)
Observations	562	562	541

Notes: Sample: Hindu households, district-level. Source: EUS 2011-12 (caste fractionalization index); HCES 2023-24 (diet proxy). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The fractionalization index is one minus the sum of squared shares of the four NSS caste groups (Scheduled Tribes, Scheduled Castes, Other Backward Classes, and Others) among Hindu households, so higher values indicate a more diverse caste composition.

Table 9: Correlation of non-veg diet (caste norms) with ST population share at district-level (2011-12)

	(1) %eat(fish)	(2) %eat(fish+meat)	(3) %UC_eat(fish+meat)
ST population share	0.17**	0.23***	0.21**
	(0.07)	(0.07)	(0.08)
Observations	585	585	578

Sample: Hindu households, district-level.

Source: EUS 2011-12 (ST population share); HCES 2011-12 (diet proxy).

Tables 7 and 8 show that districts with higher caste fractionalization have statistically significantly *lower* non-vegetarian consumption, that is, stronger dietary orthodoxy, with coefficients of −0.17, −0.35, and −0.33 for the three 2011–12 diet measures and −0.38, −0.54, and −0.68 for 2023–24 (all statistically significant at 1%, except the 2011–12 fish-only measure at 10%). Dominance by a single caste group therefore does not appear to drive norm stringency. One interpretation, which we cannot test here, is that competition for status among many caste groups raises the value of visible markers of ritual purity such as vegetarianism. Part of the association may also be compositional: because the index includes Scheduled Tribe households, districts with a large ST majority are both less fractionalized and less vegetarian. Table 9 shows that higher ST population shares are associated with higher non-vegetarian consumption, reflecting the historically less vegetarian and less caste-orthodox dietary practices of tribal communities.

These findings motivate the identification strategy we set out next. Historical upper-caste population shares, strongly correlated with present-day caste norm strength, are historically determined and plausibly exogenous to contemporary gender outcomes conditional on contemporary controls.

5.3 Untouchability and Crimes against Scheduled Castes

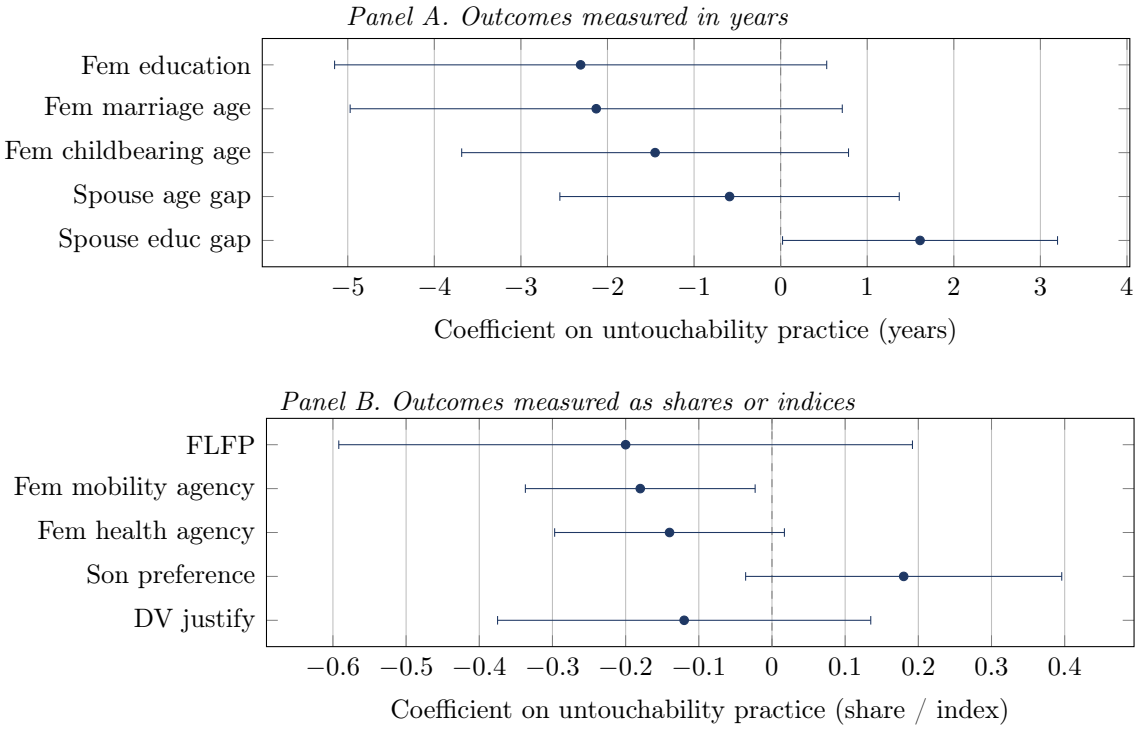
5.3.1 Untouchability

The data from the Indian Human Development Survey (IHDS) show that the illegal and highly stigmatizing practice of untouchability continues to be widely practiced, with more than 40% respondents in several states openly admitting practicing it. This is likely to be an underestimate, as respondents are less likely to admit practicing it, or view their discriminatory behaviour as natural, instead of a practice of untouchability.

Figure 4 summarises the state-level association between the IHDS untouchability practice measure and each gender outcome in a single coefficient plot. Because the outcomes are measured on two different scales, the figure has two panels: Panel A collects the outcomes denominated in years (education and the various ages and gaps) and Panel B the outcomes denominated as shares or indices (participation, agency, son preference, and DV justification). Each marker is the coefficient on untouchability practice from a

separate regression that also includes region fixed effects; the horizontal whiskers are 95% confidence intervals based on state-clustered standard errors.

Figure 4: Untouchability practice and gender norms (state level): coefficient plot



Notes. Each marker is the coefficient on the IHDS-2 (2011) untouchability practice measure from a separate state-level regression of the gender outcome on untouchability practice and region fixed effects; gender outcomes are from NFHS-4 (FLFP from EUS 2011–12). Whiskers are 95% confidence intervals constructed as the coefficient ± 1.96 *times the state-clustered standard error.* $N \approx 31$ *states.*

States with a higher prevalence of untouchability practices tend to have lower FLFPR and lower female mobility and health agency, with the agency effects statistically significant at the 5 and 10 percent levels, consistent with caste norms constraining gender outcomes. The point estimates on the year-denominated outcomes (Panel A) are also negative for education and the age outcomes, though less precisely estimated at this small state-level sample size.

5.3.2 Hate Crimes against Scheduled Castes

The deep-rooted stigmatization due to their formerly untouchable status results in violence and hate crimes against Dalits. Formal recognition of these hate crimes, or caste-based atrocities has resulted in the "Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989"[3]. The official National Crimes Record Bureau (NCRB) releases annual statistics on "Crimes in India", which records all crimes perpetrated by non-SC-ST individuals against SC-ST victims. These are compiled from the First Information

[3]https://www.ncdhr.org.in/wp-content/uploads/2026/02/Five-Years-of-Caste-Based-Atrocity-Feb-9-2026-for-web.pdf

Reports (FIRs) lodged at police stations. Like the untouchability prevalence numbers, the count of crimes against SC-ST are also likely to be underestimates, as the victims find it difficult to get their FIRs registered in view of police apathy (who carry their own caste bias and often add to the violence against the victims), as well as pressure from perpetrators preventing them from filing a report.

Table 10: Correlation of FLFP with Crimes against SCs at district-level (2019)

	(1) FLFP
Crimes against SCs (per 10,000 population)	-0.08*** (0.02)
Observations	597

Notes: FLFP is from PLFS 2019. Crimes against SCs is taken from National Crime Records Bureau Data (NCRB) 2019. SE reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Crimes against SC and gender norms, mechanism-valid states

	(1) Female education (years)	(2) Age at marriage	(3) Age at first birth	(4) FLFP	(5) DV justified	(6) Son preference	(7) Spousal age gap	(8) Spousal education gap	(9) Mobility agency	(10) Health agency
Crimes against SC (per 10,000 pop, NCRB 2019)	-0.862** (0.332)	-0.678*** (0.153)	-0.198 (0.120)	-0.131*** (0.0370)	-0.0325 (0.0561)	0.0674** (0.0262)	-0.905*** (0.282)	1.146*** (0.267)	-0.0688*** (0.0196)	-0.0514** (0.0183)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	441	441	441	434	441	441	441	441	441	441
R-squared	0.474	0.537	0.480	0.188	0.096	0.291	0.155	0.401	0.130	0.069

Notes: District-level cross-section. Gender norms from NFHS-4 (2015-16), caste norm proxy is NCRB crimes against SCs per 10,000 population (2019). Controls: district 2011 population, agricultural employment share, urban share, and log VIIRS night-light density 2012 (all from SHRUG / Census 2011). Sample excludes Bihar, West Bengal, Jammu and Kashmir, and the eight Northeast states (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura). FLFP from EUS 2011-12. Standard errors clustered at state level in parentheses. Stars: * p<0.10, ** p<0.05, *** p<0.01.

Table 10 shows a statistically significant negative association between the incidence of crimes against SCs (per 10,000 population) and district FLFPR ($\hat{\beta} = -0.08$, $p < 0.01$). Table 11 extends this to all gender outcomes in the restricted sample of 441 districts, with the same controls as the diet cross-section. Districts with more crimes against SCs, that is, stronger caste hierarchy, have less female education (-0.86 years per crime per 10,000, $p < 0.05$), earlier marriage (-0.68, $p < 0.01$), lower FLFP (-0.13, $p < 0.01$), higher son preference ($+0.07$, $p < 0.05$), a wider spousal education gap ($+1.15$ years, $p < 0.01$), a narrower spousal age gap (-0.91 years, $p < 0.01$), and lower mobility and health agency (-0.07 and -0.05, $p < 0.01$ and $p < 0.05$). This is the same pattern as for the diet proxy, including the spousal age gap, from a measure that does not depend on food taboos. The state-level untouchability measure points in the same direction for agency and FLFP but is imprecisely estimated.

6 Identification: Historical Caste Demography

The cross-sectional correlations in Section 5.1 are consistent with the view that caste norms shape gender outcomes, but they are not causal. Two concerns are paramount. First, the dietary proxy may itself be endogenous to contemporary gender norms: both

may respond jointly to unobserved factors such as urbanisation, secularisation, or developmental reach. Second, households that adopt non-vegetarian diets within a district over time may differ systematically in ways that correlate with gender outcomes, so that panel fixed-effects estimates would be hard to interpret as the effect of caste norms *per se* (Section 7.6).

Our identification strategy addresses these concerns by instrumenting for contemporary caste norms with two pre-Independence demographic variables: the 1931 Brahman population share and the 1931 Jain population share. Both communities are historically vegetarian, but the geographies of their settlement, shaped by the political, religious and commercial histories of the second millennium, are distinct. The Brahman geography tracks the historical Indo-Gangetic and Deccan political order; the Jain geography is concentrated in Western India (particularly Gujarat and Rajasthan), with outposts in Karnataka, Maharashtra, and Tamil Nadu. The two instruments are therefore non-redundant: they identify variation in caste orthodoxy from partially distinct sets of complier districts, and their non-overlapping geographies allow us to combine them in an overidentified LIML specification and test the overidentifying restriction with the Hansen J statistic.

6.1 The 1931 Brahman and Jain Shares

The Brahman population share comes from Castelló-Climent et al. (2017), originally compiled from the 1931 India Census. The data are reported at 2001 Census district boundaries, covering approximately 539 districts across 20 major states. To map these to current 2011 NFHS districts, we use the 1951-2011 district proliferation crosswalk (Harvard Dataverse, 2026), restricted to the 2001-2011 step.

For 2001 districts that split into multiple 2011 districts, we replicate the parent district's Brahman share to all child districts; for the small number of consolidations, we average across parents (no population weights are available in the crosswalk). Final coverage: **539 of 640 districts** in the analysis dataset.

The geographic logic of the instrument is visible in Figure 5. The 1931 Brahman share is highest in the same North-Central belt where contemporary dietary orthodoxy is strongest (Figure 1), and lowest in the South and East where orthodoxy has weakened most. Comparing the two maps, separated by nine decades, a partition, independence, the abolition of untouchability, and decades of affirmative action, makes the cultural persistence claim concrete: the institutional embeddedness of caste norms appears to have outlasted the major legal and political ruptures of the twentieth century.

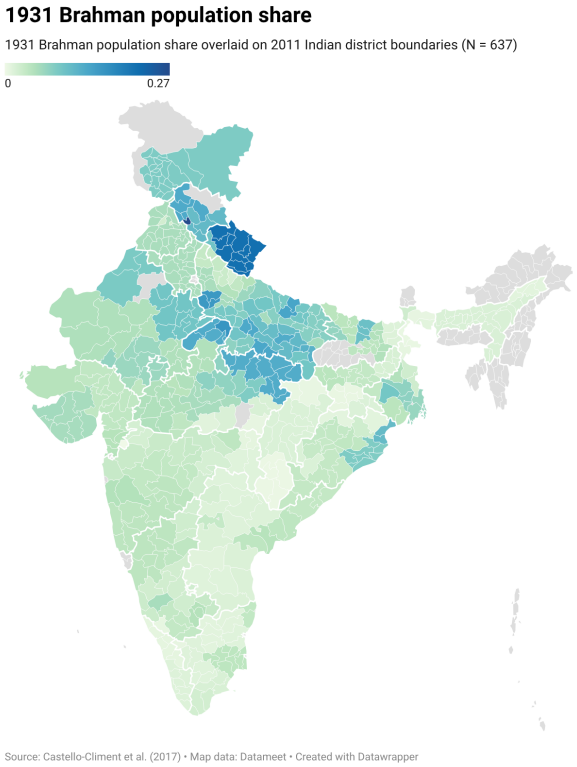


Figure 5: 1931 Brahman population share overlaid on 2011 district boundaries (Castelló-Climent et al., 2017; $N = 637$). High Brahman shares in the North reflect historical patterns of upper-caste settlement shaped by Indo-Gangetic political and religious orders. Comparing this map to Figure 1 shows that the geography of Brahmanical settlement in 1931 tracks the geography of caste dietary orthodoxy in 2011–12, the institutional persistence that motivates our identification strategy.

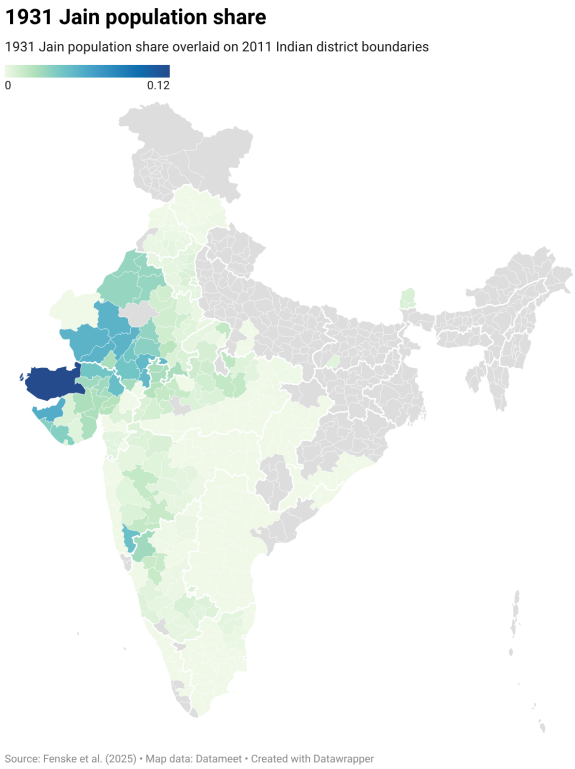


Figure 6: 1931 Jain population share overlaid on 2011 district boundaries (Fenske et al., 2025; 1931 Census, chained crosswalk). Jain settlement is densest in the commercial regions of western and central India (Gujarat, Rajasthan, and Maharashtra), with smaller concentrations in Karnataka and Tamil Nadu, and is largely absent from the eastern Gangetic plain, the Northeast, and Jammu & Kashmir. This western, mercantile, vegetarian geography is distinct from the Indo-Gangetic Brahman geography of Figure 5, which is why the two instruments capture independent variation in historical caste structure.

Jain share (1931). The Jain population share is taken from Fenske et al. (2025), originally constructed from the 1931 Census, which records the Jain population share at 1931 administrative boundaries (provinces, princely states, and British districts). Because 1931 boundaries differ substantially from current ones, we chain three crosswalks provided by Harvard Dataverse (2026) (district evolution 1872-1941, district consolidation 1941-1951, and district proliferation 1951-2024) to map to 2011 districts.

At each step, splits replicate the parent's share to children, and consolidations take the mean across parents (no population weights available in any of the crosswalks). After applying NFHS naming conventions, we obtain coverage for **321 districts across 19 states**, concentrated in Western, Central, and Southern India. Coverage is missing for the Northeast, Jammu & Kashmir, and parts of Eastern India because the 1931 Census either did not enumerate Jain populations there or the territory is now in Pakistan.

Figure 6 maps the 1931 Jain share across districts. Its geography is visibly distinct from the Brahman share in Figure 5: Jain concentrations are densest in the commercial west (Gujarat, Rajasthan, and Maharashtra) and thin out toward the east, the same fish-eating belt where the diet proxy ceases to track caste orthodoxy. This non-overlap is what makes

the two instruments jointly informative.

The two instruments have a sample correlation of approximately 0.09. This is low enough that they capture independent variation in the historical caste structure, which is essential for the overidentification analysis below.

6.2 Exclusion Restriction

The exclusion restriction is plausible for three reasons. First, the 1931 census predates Indian independence (1947) and the major policy interventions targeting caste inequality (reservations, legal prohibition of untouchability), making it unlikely to directly affect gender outcomes nearly a century later except through cultural persistence. Second, we control for contemporary district population, urbanization, agricultural employment share, log night-light density (SHRUG VIIRS 2012), and the Scheduled Tribe population share (Census 2011), which absorb the most obvious confounders. Third, the instruments measure Hindu caste demography specifically, rather than the size of Muslim, Christian, or Sikh populations, which follow different dietary norms. Scheduled Tribe populations, who have distinct cultural practices and whose presence weakens caste norms among Hindu neighbours (Agte and Bernhardt, 2024), are controlled for directly through the ST share. Together these features help isolate the effect of Hindu upper-caste orthodoxy.

These controls are necessary but, we acknowledge, not self-evidently sufficient, and the credibility of the design rests on the exclusion restriction far more than on any first-stage statistic. The most serious threat is that pre-Independence caste demography reaches contemporary gender outcomes through a channel other than caste-dietary orthodoxy. Two such channels deserve to be named explicitly. The first is historical human capital. Brahman communities were the literate caste, and the very 1931 Brahman share we use has elsewhere been tied to long-run education and prosperity (Castelló-Climent et al., 2017). If high-Brahman districts inherited deeper schooling traditions, the Brahman instrument could raise female education directly rather than through the loosening of caste-diet norms. The second is commercial development. Jains are a historically mercantile community, so the 1931 Jain share may proxy for trade, wealth, and urbanisation, each of which independently shapes women's outcomes. Our contemporary controls absorb the present-day footprint of these channels, but they cannot fully exclude a direct human-capital or commercial pathway operating in parallel with caste orthodoxy. We are therefore deliberately careful in two respects. First, the endogenous variable should be read as caste orthodoxy as a whole, indexed by diet rather than exhausted by it, so that the reported magnitudes are scaled by the diet measure and are best understood as the effect of the broader orthodoxy bundle. Second, the Hansen J test we report below disciplines only whether the two instruments *agree*; by construction it cannot detect a channel common to both, which is exactly the human-capital and commercial-prosperity

concern. We return to this limitation in interpreting the magnitudes (Section 7.3).

6.3 Estimation Sample

Using both instruments restricts the estimation sample to districts for which both the 1931 Brahman and the 1931 Jain shares are observed: the intersection of the two coverage maps, roughly 294 districts. This intersection drops the eastern Gangetic plain (Bihar, West Bengal), Jammu & Kashmir, and the Northeast. This may look like a loss of generality, but the intersection sample is the set of districts in which our endogenous variable measures what we intend it to measure, and the regions it excludes are, with near-perfect overlap, the regions in which it does not.

The reason is that the diet proxy indexes caste orthodoxy only where the caste hierarchy is organized around Brahmanical vegetarian purity. In the eastern fish-eating belt, where Bengali and Maithil Brahmans conventionally eat fish and meat, and in Kashmir, where Pandits eat meat, upper-caste status does *not* map onto vegetarianism. There a high Brahman share neither predicts low non-vegetarian consumption (the first stage fails) nor plausibly affects gender outcomes *through* dietary orthodoxy (the exclusion restriction fails); the diet measure reflects regional cuisine and distinct cultural complexes rather than the caste-purity institution we study.

The data-driven exclusion and the mechanism-driven exclusion coincide, and for a substantive reason rather than by accident. Jain mercantile settlement concentrated in the commercial, vegetarian west and centre of the subcontinent and was largely absent from the eastern fish-eating belt, so the 1931 Census records few or no Jains there. The same cultural geography that left the east outside the Brahmanical-purity complex also left it outside the Jain settlement frontier. The districts for which both historical shares are observed are therefore not a random subsample of India; they are the historical caste-purity belt itself, where Jain settlement and Brahmanical vegetarianism co-located. The regions absent from our data are absent for the same reason the mechanism does not operate in them.

We are accordingly explicit about external validity. Our estimates identify the effect of caste-dietary orthodoxy on gender outcomes *within* the caste-purity belt of northern, western, central, and southern India,[4] the population for which the question "what happens to gender norms when caste orthodoxy weakens?" is well posed, because it is the population in which caste orthodoxy is an enforced, measurable institution. We make

[4]The LIML estimation sample covers 15 states: Andhra Pradesh, Chhattisgarh, Gujarat, Haryana, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Rajasthan, Tamil Nadu, Telangana, and Uttar Pradesh. Sample sizes fall from the coverage figures above (539 Brahman and 321 Jain districts) to the estimation samples (521, 314, and 294 districts) because each specification also requires non-missing outcomes and controls, and because the island territories are dropped.

no claim for Bengal, Bihar, Kashmir, or the Northeast, where the dietary index does not capture caste orthodoxy in the first place. A natural worry is that our results are an artefact of this sample restriction, that we obtain them only because we have dropped the regions where the diet–gender relationship is weak or absent. Appendix B addresses this worry by holding the instrument fixed and varying only the sample. Holding the instrument fixed at the Brahman share and moving from the full Brahman sample (521 districts) to the LIML intersection sample (294) leaves the spousal education gap estimate nearly unchanged (-4.03 and -3.97). The female education estimate moves from -1.09 to $+1.41$, but neither is statistically significant. The gap result does not depend on the sample restriction; the education result is not pinned down by the Brahman instrument alone.

6.4 Empirical Specification

For each diet-based caste norm proxy D_d in district d and each gender norm outcome Y_d, we estimate:

$$\text{First stage:} \quad D_d = \gamma \cdot Z_d + X_d'\delta + \varepsilon_d \tag{6.1}$$

$$\text{Second stage:} \quad Y_d = \beta \cdot \widehat{D}_d + X_d'\theta + u_d \tag{6.2}$$

where Z_d is the instrument (1931 Brahman share, 1931 Jain share, or both jointly), X_d is the vector of controls, and standard errors are clustered at the state level. The exclusion restriction is that the historical Brahman or Jain population share affects contemporary gender norms only through its effect on the caste-norm diet proxy.

With two instruments and one endogenous variable, the model is over-identified, allowing two additional checks:

1. **LIML estimator**: under weak instruments, 2SLS is biased toward OLS, while LIML is approximately median-unbiased. When LIML and 2SLS estimates differ substantially, this signals a weak-instrument concern.
2. **Hansen J over-identification test**: under the null of valid instruments, the overidentifying moment conditions should hold. Large p-values support instrument exogeneity; small p-values suggest at least one instrument violates the exclusion restriction (or that the model is misspecified).

We report three complementary specifications. They differ deliberately in both the instrument used and the sample it implies; the comparison in Section B is constructed to separate the role of the sample from that of the instrument.

- **Brahman-only 2SLS** (full Brahman sample, $N = 521$ districts). The natural

starting point given the historical literature on upper-caste persistence. Its first-stage F-statistic is 8.2–9.9 depending on the dietary proxy, below the Stock–Yogo 10% critical value of 16.4 (Stock and Yogo, 2005). The Brahman share alone is thus a *weak* instrument: its first stage is correctly signed and statistically significant throughout the sample, but not strong enough to pin down the second-stage coefficient, which is why we treat its point estimates as imprecise and introduce a second instrument.

- **Jain-only 2SLS** (19 major states, $N = 314$ districts). The Jain share is observed in fewer states, but where it is observed it is a stronger instrument, with a first-stage F-statistic of 17.5 and 22.5 for the two broader dietary proxies, above the Stock–Yogo 10% critical value, and 10.0 for the fish-only proxy. Its sample is almost exactly the intersection sample used by the LIML specification below.

- **LIML with both instruments jointly (preferred)** (intersection sample, $N =$ 294 districts in 15 states; 288 for the upper-caste proxy). Because the Jain-only and LIML samples nearly coincide, moving from Jain-only to LIML adds the Brahman instrument and an overidentifying restriction while holding the estimation population nearly fixed. LIML is approximately median-unbiased under weak instruments; the joint first-stage F-statistic is 19.6 and 24.8 for the two broader proxies and 8.7 for the fish-only proxy, against a Stock–Yogo critical value of 8.68 for LIML with two instruments (10% maximal size, computed under homoskedastic errors); and the overidentifying restriction lets us test whether the two instruments, whose compliers differ across geography, deliver consistent estimates (Hansen J).

Standard errors are clustered by state. With 15 state clusters in the LIML sample and 19 in the Jain-only sample, clustered standard errors may understate sampling uncertainty, so we treat p-values near conventional thresholds with caution.

7 Causal Estimates

This section reports the causal effect of caste norms on gender outcomes. We begin with the first-stage relationships between the 1931 Brahman and Jain shares and the contemporary dietary proxies, then present the overidentified LIML specification, interpret the magnitudes, and examine the marriage-market mechanisms. We close with robustness checks and heterogeneity by economic development. The single-instrument 2SLS estimates are reported in Appendix C, and a comparison that separates the role of the sample from that of the instrument is in Appendix B.

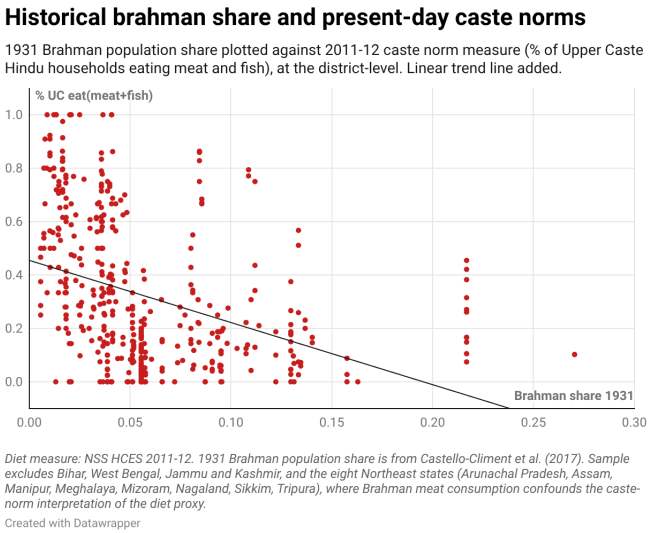


Figure 7: Scatter of the 1931 Brahman population share (x-axis) against the district-level share of upper-caste Hindu households consuming meat and fish in 2011–12 (y-axis), with a linear fit. Each point is a district. The sample is restricted to the mechanism-valid states, excluding Bihar, West Bengal, Jammu & Kashmir, and the eight Northeast states, where upper-caste meat consumption confounds the caste-norm interpretation of the diet proxy. The negative slope (a higher 1931 Brahman share predicts lower upper-caste non-vegetarian consumption nine decades later) is the visual analogue of the first-stage regression. Brahman share from Castelló-Climent et al. (2017); diet from NSS HCES 2011–12.

7.1 First Stage

Figure 7 provides the visual first stage on the mechanism-valid sample: across these districts, a higher 1931 Brahman population share predicts substantially lower upper-caste non-vegetarian consumption nine decades later, and the relationship is approximately linear. The formal first-stage regressions in Tables 12–15 show that this relationship holds conditional on contemporary population, urbanisation, agricultural employment, economic activity, and the Scheduled Tribe population share.

Table 12: First Stage: 1931 Brahman Share as Instrument for Caste Norms

	(1) %eat(fish)	(2) %eat(fish+meat)	(3) %UC_eat(fish+meat)
Brahman population share 1931	-1.871***	-2.554***	-2.412***
	(0.655)	(0.810)	(0.794)
Controls	Yes	Yes	Yes
Observations	522	522	515
R-squared	0.375	0.355	0.320
First-stage F-stat	8.16	9.94	9.24

Dependent variable: district-level diet-based caste norm proxy from HCES 2011-12 (NSS 68th round).
Instrument: Brahman population share from 1931 Census, mapped to present-day districts.
Controls included: district population (Census 2011), agricultural employment share, urban share, scheduled tribe population share, and log night light density (SHRUG VIIRS) in 2012. Standard errors clustered at state level in parentheses.

Tables 12 and 13 report the first-stage relationship between the 1931 Brahman population share and contemporary non-vegetarian consumption, for HCES 2011–12 and HCES

Table 13: First Stage: 1931 Brahman Share as Instrument for Caste Norms

	(1) %eat(fish)	(2) %eat(fish+meat)	(3) %UC_eat(fish+meat)
Brahman population share 1931	-1.957***	-2.673***	-2.614**
	(0.615)	(0.928)	(1.101)
Controls	Yes	Yes	Yes
Observations	507	507	489
R-squared	0.333	0.290	0.265
First-stage F-stat	10.13	8.30	5.64

Dependent variable: district-level diet-based caste norm proxy from HCES 2023-24.
Instrument: Brahman population share from 1931 Census, taken from Castelló-Climent et al. (2017).
Controls included: district population (Census 2011), agricultural employment share, urban share, scheduled tribe population share, and log night light density (SHRUG VIIRS) in 2012. Standard errors clustered at state level in parentheses.

2023–24 respectively. Across all three dietary proxies (fish among Hindus, fish and meat among Hindus, and fish and meat among upper-caste Hindus), the coefficient on the 1931 Brahman share is negative and statistically significant at 1% or 5%, consistent with cultural persistence of Brahmanical vegetarianism. The first-stage F-statistic ranges from 8.2 to 9.9 in 2011–12 and from 5.6 to 10.1 in 2023–24; these values are below the Stock–Yogo 10% critical value of 16.4, motivating the introduction of a second instrument.

Table 14: First Stage: 1931 Jain Share as Instrument for Caste Norms

	(1) %eat(fish)	(2) %eat(fish+meat)	(3) %UC_eat(fish+meat)
Jain population share 1931	-3.639***	-7.379***	-6.646***
	(1.153)	(1.764)	(1.402)
Controls	Yes	Yes	Yes
Observations	314	314	308
R-squared	0.260	0.251	0.203
First-stage F-stat	9.96	17.50	22.46

Dependent variable: district-level diet-based caste norm proxy from HCES 2011-12 (NSS 68th round).
Instrument: Jain population share from 1931 Census, mapped to present-day districts via chained crosswalks (1931→1941→1951→2011).
Controls included: district population (Census 2011), agricultural employment share, urban share, scheduled tribe population share, and log night light density (SHRUG VIIRS) in 2012. Standard errors clustered at state level in parentheses.

Table 14 reports the first stage for the 1931 Jain population share. The coefficient is negative, large in magnitude, and statistically significant at 1% for all three dietary proxies; the first-stage F-statistic is 17.5 and 22.5 for the two broader proxies, above the Stock–Yogo 10% critical value of 16.4, and 10.0 for the fish-only proxy. This is consistent with the dense settlement of Jains in historically commercial regions of Western India that became centres of strict vegetarianism.

Table 15 reports the joint first stage. Both instruments remain statistically significant

Table 15: First Stage: 1931 Brahman and Jain Shares as Joint Instruments

	(1) %eat(fish)	(2) %eat(fish+meat)	(3) %UC_eat(fish+meat)
Brahman population share 1931	-1.226**	-2.854***	-2.583***
	(0.479)	(0.790)	(0.774)
Jain population share 1931	-3.579***	-7.297***	-6.666***
	(0.910)	(1.269)	(1.036)
Controls	Yes	Yes	Yes
Observations	294	294	288
R-squared	0.412	0.482	0.388
Joint F-stat (both IVs)	8.71	19.59	24.78

Dependent variable: district-level diet-based caste norm proxy from HCES 2011-12 (NSS 68th round).
Instruments: Brahman and Jain population shares from 1931 Census, mapped to present-day districts.
Joint F-stat from test of both instruments being zero.
Controls included: district population (Census 2011), agricultural employment share, urban share, scheduled tribe population share, and log night light density (SHRUG VIIRS) in 2012. Standard errors clustered at state level in parentheses.

predictors of contemporary dietary practices when entered simultaneously, indicating that they identify partially distinct variation in historical caste demography. The joint F-statistic on the two excluded instruments is 19.6 for fish and meat among Hindus, 24.8 for fish and meat among upper-caste Hindus, and 8.7 for the fish-only measure. For the two broader measures this is an improvement on either instrument used alone; for the fish-only measure it is not.

The single-instrument 2SLS estimates are reported in Appendix C. The Brahman-only estimates are imprecise, reflecting the weak first stage: they statistically significantly narrow the spousal education gap and widen the spousal age gap, but the coefficient on female education is negative and not statistically significant. The Jain-only estimates, whose first stage is stronger, are close to the LIML estimates below.

7.2 LIML with Both Instruments (Preferred)

Table 16 reports our preferred specification: LIML using both instruments jointly. LIML is chosen over 2SLS because it is approximately median-unbiased under weak instruments; the overidentified structure permits a test of the overidentifying restriction with the Hansen J statistic. We report all three dietary proxies in Panels A–C of the table; the narrative follows Panel B (share of Hindu households consuming fish or meat), with qualitatively similar results in the other panels. All specifications control for district population, agricultural employment share, urban share, the Scheduled Tribe population share, and log night-light density. With the Scheduled Tribe share included, the headline estimates remain large and statistically significant.

The headline results are as follows:

Table 16: IV Estimates: Effect of Caste Norms on Gender Norms (LIML, Overidentified)

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	8.230**	2.923	0.120	0.503	1.393***	-0.308	9.964***	-8.803***	0.150	0.013
	(3.451)	(2.243)	(1.371)	(0.657)	(0.467)	(0.184)	(2.982)	(2.451)	(0.119)	(0.155)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71
Hansen J p-value	0.403	0.923	0.888	0.432	0.202	0.778	0.632	0.911	0.952	0.479
Observations	294	294	294	287	294	294	294	294	294	294
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	3.438**	1.339	0.051	0.178	0.634***	-0.142*	4.572***	-4.059***	0.069	0.004
	(1.415)	(1.027)	(0.659)	(0.220)	(0.126)	(0.071)	(0.672)	(0.638)	(0.058)	(0.072)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	19.59	19.59	19.59	19.59	19.59	19.59	19.59	19.59	19.59	19.59
Hansen J p-value	0.164	0.834	0.878	0.387	0.491	0.780	0.453	0.813	0.989	0.453
Observations	294	294	294	287	294	294	294	294	294	294
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	3.834**	1.468	0.062	0.185	0.669***	-0.155*	4.879***	-4.398***	0.071	-0.001
	(1.637)	(1.139)	(0.734)	(0.249)	(0.146)	(0.078)	(0.802)	(0.771)	(0.066)	(0.083)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	24.78	24.78	24.78	24.78	24.78	24.78	24.78	24.78	24.78	24.78
Hansen J p-value	0.145	0.832	0.880	0.386	0.542	0.790	0.431	0.785	0.940	0.426
Observations	288	288	288	281	288	288	288	288	288	288

LIML estimates with two instruments (overidentified model, less biased than 2SLS under weak IVs). Sample: Districts with non-missing 1931 Brahman and Jain shares. Gender norms from NFHS-4 (2015-16), collapsed to district level, matched with caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). Instruments are 1931 Brahman share (Castelló-Climent et al., 2017) and 1931 Jain population share (Census 1931, chained crosswalk to 2011 districts). Controls included are district population (Census 2011), agricultural employment share, urban share, and scheduled tribe population share in 2011, and log night light density (SHRUG VIIRS, 2012) in 2012. FLFP is from PLFS 2011-12. First-stage F-stat is from the joint test of both instruments. Hansen J p-value tests instrument exogeneity (H0: overidentifying restrictions are valid); large values support instrument validity. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

Table 17: IV Estimates: Effect of Caste Norms on Marriage Ages (LIML, South Control)

	Panel A: %eat(fish)		Panel B: %eat(fish+meat)		Panel C: %UC_eat(fish+meat)	
	Wife Marr. Age	Husband Marr. Age	Wife Marr. Age	Husband Marr. Age	Wife Marr. Age	Husband Marr. Age
Caste norm (endogenous)	5.328	9.465*	3.758**	6.702**	3.694**	6.613**
	(3.080)	(4.999)	(1.632)	(2.652)	(1.616)	(2.719)
Observations	294	294	294	294	288	288
First-stage F	8.7	8.7	19.6	19.6	24.8	24.8
Hansen J (p-value)	0.898	0.901	0.757	0.963	0.817	0.898

LIML estimates with two instruments (overidentified). Endogenous variable is the district caste norm (diet proxy); instruments are 1931 Brahman and Jain population shares. *Wife marriage age*: age at first cohabitation (NFHS-4, v511). *Husband marriage age*: husband's current age minus years since first cohabitation (NFHS-4 v730 − v512); non-missing only for co-resident couples. Controls included: district population (Census 2011), agricultural employment share, urban share, scheduled tribe population share in 2011, log night light density (SHRUG VIIRS, 2012), and a South India dummy (=1 for Kerala, Karnataka, Tamil Nadu, Andhra Pradesh, Telangana). No other region dummies are included. First-stage F is the joint F-statistic from the main LIML spec (Table 18): OLS of the caste norm on both instruments and base controls, testing joint instrument significance. Standard errors clustered at state level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

1. **Spousal education gap (our flagship result).** Weaker caste norms close the spousal education gap by 4.06 years ($p < 0.01$). This is the most stable finding in the paper across instruments and proxies. Section 7.4 examines the mechanism.

2. **Female education.** Weaker caste norms raise average female education by +3.44 years ($p < 0.05$). This magnitude sits between the imprecise (weak-instrument) Brahman-only point estimate (−1.09) and the Jain-only point estimate (+5.27 years). Appendix B examines whether the difference between the two single-instrument estimates reflects the sample or the instrument.

3. **Son preference.** Weaker caste norms reduce son preference by 0.14 ($p < 0.10$). The district mean of son preference in the estimation sample is 0.134 (Table A10), so the full-range effect is about the size of the mean; an interquartile shift reduces son preference by 0.065, about half the mean.

4. **Spousal age gap.** Weaker caste norms are associated with a *wider* spousal age gap, by +4.57 years ($p < 0.01$). The decomposition in Table 17 shows that this reflects a larger rise in men's age at marriage than in women's. We interpret it in Section 7.4 as consistent with a marriage-market channel.

5. **DV justification.** Weaker caste norms *raise* stated justification of domestic violence by +0.63 ($p < 0.01$). This is the other counterintuitive result. The DV justification question may capture a respondent's willingness to report acceptance of a stigmatised attitude rather than the attitude itself. Experienced violence also rises, and we discuss both in Section 7.5.

6. **Female labour force participation.** The coefficient is +0.18 and statistically indistinguishable from zero. We emphasise this null and return to its interpretation in Section 8.3.

7. **Female agency, marriage age, first birth age.** Coefficients are positive but imprecisely estimated and do not reach conventional levels of statistical significance.

The Hansen J test does not reject the overidentifying restriction for any outcome in any panel (p-values 0.15 to 0.99): two instruments whose compliers differ across geography are statistically consistent with a common effect, which is what our argument requires. We read this as supportive of the exclusion restriction, with one caveat we state plainly. The Hansen J tests only whether the two instruments imply the *same* effect, so it cannot detect a violation common to both. Because both instruments proxy features of historical caste demography, a channel running from pre-Independence caste structure to contemporary gender outcomes other than through dietary orthodoxy (historical literacy or commercial prosperity, say) would bias both in the same direction and leave J undisturbed. We address this shared-channel concern not through J but through our contemporary controls

(population, urbanisation, agricultural employment, the Scheduled Tribe share, night-light density), which absorb the most obvious development confounders.

7.3 Magnitudes

The coefficients in Table 16 measure the effect of a change in the diet share from zero (no Hindu household eats fish or meat) to one (every Hindu household does), a range far wider than the variation across districts. Table A9 rescales them to an interquartile shift in each proxy. For our preferred fish-and-meat proxy, the interquartile range in the LIML sample is 0.46 (from 0.11 to 0.56). An interquartile weakening of caste dietary orthodoxy narrows the spousal education gap by 1.85 years, raises female education by 1.57 years, widens the spousal age gap by 2.09 years, and reduces son preference by 0.065. The corresponding sample means are 1.54, 7.03, 4.99, and 0.134 (Table A10). These are large effects: the implied change in the education gap exceeds its sample mean, which suggests caution in extrapolating the linear specification across the full interquartile range.

The outcomes on which our IV delivers precisely estimated effects (female education, the spousal education gap, and son preference) are pre-labour-market outcomes that describe whom women marry and the human capital they bring to the market. Labour supply, conditional on these, does not respond detectably. This is consistent with caste norms operating upstream of labour market choices: they constrain the human capital and marriage-market options available to women, and labour supply decisions are made within those constraints.

7.4 Marriage-Market Mechanisms

Weaker caste norms narrow the spousal education gap by 4.06 years in the LIML specification. To our knowledge, no earlier study estimates the effect of caste norms on spousal education gaps. The mechanism we propose is caste endogamy (the norm of marrying within caste), which shrinks the pool of potential spouses available to an educated woman. Where women have been less educated than men, endogamous marriage generates large spousal education gaps that can persist even as women's education rises; loosening endogamy allows the marriage market to match more closely on education. We do not observe inter-caste marriage directly, so the estimates are consistent with this mechanism rather than a test of it.

Table 18 decomposes the gap on the sample of couples for which both spouses' education is observed. Weaker caste norms raise wives' education by 3.54 years ($p < 0.05$, fish-and-meat proxy) and have no statistically significant effect on husbands education (-0.68, standard error 1.24). On this sample the gap estimate is identical to the headline estimate (-4.06). The gap therefore narrows because wives are more educated, not because women marry less-educated men. The decomposition cannot by itself separate the endogamy

Table 18: Husband and Wife Education on the Couple Sample (LIML)

	Panel A: Fish			Panel B: Fish or meat			Panel C: UC fish or meat		
	(1) Husband educ	(2) Wife educ	(3) Gap	(4) Husband educ	(5) Wife educ	(6) Gap	(7) Husband educ	(8) Wife educ	(9) Gap
Diet proxy	-1.822	8.385**	-8.803***	-0.675	3.544**	-4.059***	-0.828	3.952**	-4.398***
	(3.707)	(3.761)	(2.451)	(1.235)	(1.586)	(0.638)	(1.490)	(1.802)	(0.771)
Observations	294	294	294	294	294	294	288	288	288
First-stage F-stat	8.71	8.71	8.71	19.59	19.59	19.59	24.78	24.78	24.78
Hansen J p-value	0.138	0.469	0.911	0.173	0.237	0.813	0.174	0.216	0.785

Notes: LIML estimates, same specification as Table 16: instruments are 1931 Brahman and Jain population shares; controls are district population, agricultural employment share, urban share, ST share and log night light density 2012; standard errors clustered by state in parentheses. Outcomes are district means from NFHS-4 (2015–16) over women for whom both own and husband's years of education are reported: husband's years of education, wife's years of education, and their difference (husband minus wife). First-stage F-stat is the joint test of both instruments from the first stage in Table 16. Hansen J p-value tests the overidentifying restriction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

channel from a direct effect of caste norms on women's schooling: relaxed endogamy could raise the return to educating daughters by widening the pool of educated grooms, or caste norms could restrict girls' schooling independently of marriage.

For the spousal age gap, the decomposition in Table 17 points to a marriage-market channel. Splitting the gap into its two components, weaker caste norms raise women's age at marriage, as our framework predicts (+3.76 years, $p < 0.05$, fish-and-meat proxy), but raise men's age at marriage by almost twice as much (+6.70 years, $p < 0.05$). The gap widens because grooms marry later, while brides also marry later but by less.

We read this as consistent with a change in how marriages are underwritten. Under strong caste orthodoxy, marriage is arranged within the caste and underwritten by kin: the groom does not need independent economic standing, and both spouses marry at customary ages. As that institution weakens, marriage comes to rest on the individual, and the groom must demonstrate his own earning capacity, which accrues with age, so men's age at marriage rises steeply. Women's age at marriage rises too, but by less, because it remains bounded by fertility norms. An alternative reading, that relaxed endogamy opens the market to inter-caste and love marriages drawn from a wider age pool, fits the decomposition less well: a wider pool of partners chosen by the spouses themselves would, if anything, predict a smaller age gap.

7.5 Domestic Violence

For DV justification, the dependent variable is a self-reported attitude about whether physical discipline of a wife is acceptable in specified circumstances. Attitudes of this kind plausibly carry social desirability bias that varies with caste orthodoxy: respondents in caste-orthodox districts may be less willing to voluntarily state an acceptance of violence, regardless of the underlying attitude or practice, so that a measured *increase* in stated justification where caste norms are weaker may reflect greater candor rather than greater

Table 19: IV Estimates: Effect of Caste Norms on Experienced Domestic Violence (LIML)

	All women		DV-module women	
	(1) Experienced DV	(2) Experienced IPV	(3) Experienced DV	(4) Experienced IPV
Panel A: Share of Hindu HHs eating fish				
Diet proxy	0.064*	0.063*	0.437*	0.485*
	(0.035)	(0.034)	(0.230)	(0.256)
First-stage F-stat	8.71	8.71	8.71	8.71
Hansen J p-value	0.167	0.169	0.740	0.794
Observations	294	294	294	294
Outcome mean	0.028	0.028	0.243	0.285
Panel B: Share of Hindu HHs eating fish or meat				
Diet proxy	0.029**	0.029**	0.202*	0.224*
	(0.013)	(0.013)	(0.095)	(0.108)
First-stage F-stat	19.59	19.59	19.59	19.59
Hansen J p-value	0.475	0.476	0.901	0.934
Observations	294	294	294	294
Outcome mean	0.028	0.028	0.243	0.285
Panel C: Share of upper-caste Hindu HHs eating fish or meat				
Diet proxy	0.029**	0.029**	0.210*	0.236*
	(0.013)	(0.013)	(0.102)	(0.117)
First-stage F-stat	24.78	24.78	24.78	24.78
Hansen J p-value	0.521	0.519	0.896	0.922
Observations	288	288	288	288
Outcome mean	0.028	0.028	0.241	0.283

Notes: LIML estimates, same specification as Table 16: instruments are 1931 Brahman and Jain population shares; controls are district population, agricultural employment share, urban share, ST share and log night light density 2012; standard errors clustered by state in parentheses. Outcomes are district shares of women from NFHS-4 (2015–16) who report physical or sexual violence by a partner (IPV) or by a partner or other household member (DV). Columns (1) and (2) use the district means in the analysis data, which average over all women and code women not selected for the domestic violence module (about 88% of women) as zero. Columns (3) and (4) average over women selected and interviewed for the module (column 4: ever-partnered women only). First-stage F-stat is the joint test of both instruments from the first stage in Table 16. Outcome mean is the unweighted district mean in the estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

acceptance. The coefficient on *experienced* domestic violence, however, is also positive and statistically significant (Table 19): 0.029 ($p < 0.05$) on the district measure, whose mean is 0.028 because NFHS codes women outside the domestic violence module as zero, and 0.20 ($p < 0.10$) when restricted to women in the module, against a module mean of 0.24. Because experienced violence is also self-reported, this is consistent with greater candor in districts with weaker caste norms, but it is equally consistent with a real increase in violence, and our data cannot distinguish the two. We emphasize that this is an interpretation we cannot test directly with these data, not a result; disciplining the reporting margin properly would require a measurement design we do not have here, and we flag it as a limitation.

7.6 Robustness

The single-instrument estimates (Appendix C) and the comparison in Appendix B, which holds the sample fixed while varying the instrument, show how the preferred estimates relate to their building blocks. On the common LIML sample, the Brahman-only estimates of the spousal education gap (−3.97) and female education (+1.41, not statistically significant) have 95% confidence intervals that overlap with the LIML estimates, and OLS on the same sample gives a smaller gap coefficient (−2.43) than LIML (−4.06).

A natural follow-up question is whether the cross-sectional relationships line up with within-district changes in dietary practices over time. Appendix Table A5 compares OLS estimates with and without district fixed effects. The cross-sectional estimates (columns 1, 3, 5) show that districts with higher non-vegetarian diet shares have better gender outcomes. However, when we include district fixed effects (columns 2, 4, 6), several coefficients flip sign or attenuate toward zero.

This pattern suggests that cross-district and within-district variation capture different margins. We view the cross-sectional margin as the policy-relevant one, and it is the margin our instruments use. The IV captures the effect of persistent, historically-rooted differences in caste orthodoxy across districts: the kind of deep structural variation that matters for long-run policy and that our framework holds to be sticky. The within-district fixed-effects estimator, by contrast, captures short-run dietary change within a district, which reflects compositional shifts (who adopts non-vegetarian diets first as incomes rise) and development confounds (urbanization, migration, sectoral change). The sign reversals between the two are consistent with a story in which the marginal adopters of non-vegetarian diet *within* a district over a twelve-year window differ systematically from the populations that characterize districts with persistently weaker caste norms.

The policy-relevant question is whether shifting a district's caste norms to resemble those of a district with a different historical profile would improve gender outcomes. The cross-

sectional IV is the estimator that speaks to this question.

7.7 Heterogeneity by Economic Development

The preceding evidence suggests that weaker caste norms are associated with better gender outcomes on average. We now ask whether this relationship varies with the level of economic development. We use two district-level proxies for development: night-light density from satellite imagery (SHRUG VIIRS 2012) and mean per-capita expenditure (MPCE, from the EUS 2011–12). If caste norms bind primarily in traditional economies, their association with gender outcomes should weaken in more developed districts, where other forces such as education, urbanisation, and exposure to wider labour markets drive gender-norm change. We test this by interacting each caste-norm proxy with a dummy for above-median development; the main effect then captures the association in less-developed districts and the interaction the additional effect in more-developed ones.

Table 20: Heterogeneous Effects of Caste Norms by Economic Activity

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	0.195	0.530	0.123	-0.109	0.175**	-0.018	1.926***	-1.534***	0.062*	0.038
	(0.709)	(0.439)	(0.339)	(0.090)	(0.075)	(0.051)	(0.489)	(0.428)	(0.031)	(0.028)
High night lights (=1)	0.135	-0.127	-0.081	-0.062	-0.010	0.020	-0.327	0.347*	-0.019	-0.011
	(0.292)	(0.163)	(0.108)	(0.040)	(0.031)	(0.029)	(0.238)	(0.182)	(0.018)	(0.015)
%eat(fish) x High night lights	-0.278	-0.190	-0.173	0.067	-0.044	-0.062	0.486	-0.283	0.031	0.037
	(0.651)	(0.484)	(0.446)	(0.083)	(0.072)	(0.049)	(0.472)	(0.346)	(0.035)	(0.031)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	606	606	606	582	606	606	606	606	606	606
R-squared	0.466	0.474	0.414	0.084	0.095	0.213	0.253	0.342	0.106	0.072
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	0.662	0.942**	0.191	0.009	0.210***	-0.076	2.003***	-1.746***	0.105***	0.076***
	(0.582)	(0.378)	(0.259)	(0.082)	(0.059)	(0.050)	(0.468)	(0.345)	(0.023)	(0.025)
High night lights (=1)	0.454	0.232	0.110	-0.060	-0.027	0.012	-0.594**	0.270	0.001	0.015
	(0.325)	(0.200)	(0.132)	(0.041)	(0.033)	(0.033)	(0.255)	(0.233)	(0.019)	(0.017)
%eat(fish+meat) x High night lights	-0.754	-0.819*	-0.580	0.078	0.075	-0.042	1.309**	-0.238	-0.012	-0.029
	(0.567)	(0.457)	(0.361)	(0.073)	(0.069)	(0.047)	(0.555)	(0.417)	(0.033)	(0.032)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	606	606	606	582	606	606	606	606	606	606
R-squared	0.471	0.486	0.422	0.081	0.199	0.247	0.400	0.403	0.134	0.077
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	0.321	0.700**	0.147	-0.006	0.163***	-0.047	1.770***	-1.202***	0.071***	0.053**
	(0.517)	(0.293)	(0.223)	(0.070)	(0.056)	(0.048)	(0.442)	(0.365)	(0.024)	(0.022)
High night lights (=1)	0.283	0.153	0.086	-0.057	-0.028	0.022	-0.588**	0.451**	-0.009	0.006
	(0.323)	(0.169)	(0.120)	(0.038)	(0.033)	(0.031)	(0.252)	(0.206)	(0.018)	(0.015)
%UC_eat(fish+meat) x High night lights	-0.510	-0.693*	-0.539	0.065	0.060	-0.053	1.169**	-0.457	0.010	-0.009
	(0.483)	(0.361)	(0.328)	(0.066)	(0.056)	(0.045)	(0.502)	(0.338)	(0.029)	(0.027)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	591	591	591	567	591	591	591	591	591	591
R-squared	0.479	0.492	0.429	0.066	0.158	0.238	0.373	0.360	0.109	0.066

Sample: District-level cross-section. Gender norms from NFHS-4 (2015-16), collapsed to district level, matched with caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). High night lights = 1 if district log night light density is above the sample median (SHRUG VIIRS, 2012), 0 otherwise. Controls included are district population (Census 2011), agricultural employment share, and urban share in 2011. FLFP is from PLFS 2011-12. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

Tables 20 and 21 report the two sets of interactions, for night lights and MPCE respectively. We focus on Panel B (share of Hindu households eating fish or meat), with Panels A and C (the fish-only and upper-caste measures) giving a similar picture. The results carry two messages that are consistent across both proxies for development.

First, the main effects reproduce the central findings of the paper. In less-developed districts, weaker caste norms are associated with a narrower spousal education gap (-1.79 and -2.18 years under night lights and MPCE, both $p < 0.01$) and a wider spousal age gap ($+2.35$ and $+3.02$ years, both $p < 0.01$). Under night lights they are also associated with more female education ($+1.02$ years, $p < 0.10$) and later marriage ($+0.68$ years, $p < 0.10$); under MPCE the marriage-age coefficient is similar in size ($+0.58$) but not statistically significant. Agency coefficients are positive but mostly imprecise:

Table 21: Heterogeneous Effects of Caste Norms by Economic Activity (MPCE)

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	-0.141	0.590	-0.027	-0.074	0.040	-0.091	2.248***	-1.696***	0.081*	0.072*
	(0.342)	(0.360)	(0.359)	(0.070)	(0.055)	(0.064)	(0.570)	(0.406)	(0.040)	(0.038)
High MPCE (=1)	0.255	0.319	0.195	0.029	-0.062*	-0.038	-0.300	0.139	0.007	0.016
	(0.218)	(0.192)	(0.144)	(0.035)	(0.030)	(0.024)	(0.237)	(0.171)	(0.018)	(0.020)
%eat(fish) x High MPCE	0.916	0.263	0.543	0.084	0.234***	0.031	-0.315	-0.203	0.003	-0.024
	(0.761)	(0.453)	(0.399)	(0.089)	(0.058)	(0.064)	(0.471)	(0.447)	(0.046)	(0.042)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	574	574	574	566	574	574	574	574	574	574
R-squared	0.477	0.508	0.457	0.086	0.130	0.215	0.262	0.340	0.129	0.090
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	0.320	0.745*	-0.155	0.109	0.167**	-0.150**	2.777***	-2.146***	0.094***	0.059*
	(0.466)	(0.374)	(0.326)	(0.081)	(0.075)	(0.058)	(0.462)	(0.279)	(0.032)	(0.031)
High MPCE (=1)	0.355	0.364	0.095	0.082*	-0.057*	-0.064**	-0.123	-0.090	0.005	0.008
	(0.328)	(0.262)	(0.204)	(0.043)	(0.029)	(0.029)	(0.226)	(0.234)	(0.025)	(0.025)
%eat(fish+meat) x High MPCE	0.319	0.059	0.439	-0.024	0.143**	0.057	-0.369	0.330	0.011	0.006
	(0.678)	(0.460)	(0.373)	(0.083)	(0.059)	(0.053)	(0.448)	(0.458)	(0.044)	(0.040)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	574	574	574	566	574	574	574	574	574	574
R-squared	0.477	0.515	0.452	0.097	0.212	0.264	0.393	0.409	0.162	0.098
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	-0.028	0.432	-0.200	0.044	0.102*	-0.100*	2.373***	-1.507***	0.062**	0.042
	(0.374)	(0.281)	(0.277)	(0.063)	(0.052)	(0.052)	(0.401)	(0.321)	(0.030)	(0.027)
High MPCE (=1)	0.232	0.237	0.090	0.047	-0.075**	-0.038	-0.308	0.174	-0.007	0.002
	(0.271)	(0.225)	(0.181)	(0.039)	(0.030)	(0.023)	(0.258)	(0.243)	(0.022)	(0.022)
%UC_eat(fish+meat) x High MPCE	0.546	0.247	0.446	0.045	0.171***	0.009	-0.133	-0.047	0.028	0.011
	(0.586)	(0.402)	(0.336)	(0.069)	(0.050)	(0.043)	(0.450)	(0.458)	(0.039)	(0.035)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	559	559	559	551	559	559	559	559	559	559
R-squared	0.484	0.518	0.457	0.081	0.182	0.246	0.372	0.357	0.135	0.084

Sample: District-level cross-section. Gender norms from NFHS-4 (2015-16), collapsed to district level, matched with caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). High MPCE = 1 if district log mean per capita expenditure 2011 (Rs '000s) is above the sample median, 0 otherwise. MPCE constructed from PLFS 2011-12. Controls included are district population (Census 2011), agricultural employment share, and urban share in 2011. FLFP is from PLFS 2011-12. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

mobility agency is statistically significant only under MPCE (+0.06, $p < 0.10$), and health agency under neither proxy. Son preference falls under the MPCE specification (-0.15, $p < 0.05$). Consistent with the reporting interpretation we develop for the IV estimates, stated justification of domestic violence is *higher* where caste norms are weaker (+0.28 under night lights and +0.21 under MPCE, statistically significant at 1% and 5%). The coefficient on female labour force participation for the fish-and-meat measure is small and statistically indistinguishable from zero under both proxies, in line with the FLFP null that runs throughout the paper; for the fish-only measure (Panel A) it is negative and statistically significant (-0.17 and -0.13).

Second, there is little evidence that economic development substitutes for the weakening of caste norms. Across both tables, most interaction terms are not statistically significant: the association between caste norms and the spousal education gap, mobility and health agency, son preference, and marriage age does not differ detectably between more- and less-developed districts. The handful of statistically significant interactions does not form a consistent pattern. Under the night-light proxy, the education association is weaker in developed districts (interaction -0.97, $p < 0.10$), the FLFP coefficient is slightly higher there (+0.14, $p < 0.10$), and the spousal age-gap association is stronger there (interaction +1.10, $p < 0.05$). Under the MPCE proxy, the only consistent interaction is on DV justification, which is more positive in richer districts (interaction +0.16 to +0.24, statistically significant at 1% across all three panels), which is consistent with reading DV justification as partly capturing willingness to report.

We interpret these interactions as descriptive rather than causal. Neither night-light density nor MPCE is randomly assigned, and both correlate with unobserved factors such as urbanization, migration, and state capacity that independently affect gender

outcomes. Because most interactions are not statistically significant, we cannot detect that the association between caste norms and gender outcomes weakens in more developed districts. This is consistent with, though not proof of, our framing of caste as a sticky, institutionally embedded hierarchy that economic growth does not by itself dissolve.

8 Caste Norms and Female Labour Force Participation

8.1 The FLFP Puzzle

Female labour force participation in India presents an apparent paradox: as gender norms have improved substantially over two decades, FLFPR first fell and then only partially recovered. This seeming contradiction has been invoked as evidence that structural or economic factors, rather than norms, drive labour supply. Our evidence is consistent with this interpretation.

Rural FLFPR was approximately 50% in 1999–2000, a level comparable to Spain, Korea, and Greece at the time. It declined to a trough near 24–26% around 2017–18 before rising to approximately 38% by 2022–23. This U-shaped trajectory coincided with two decades of structural transformation: the share of agriculture, which had historically absorbed the bulk of female labour, fell sharply in both GDP and employment, while the period was one of "jobless growth" in which the sectors that expanded did not generate commensurate employment. The result was weak local labour demand for women: the jobs women had been doing contracted faster than accessible jobs were created elsewhere, so that the contraction shows up as falling participation rather than as reallocation across sectors.

8.2 Evidence from Companion Papers

The demand-side reading we adopt here is developed in detail in two companion papers, both of which use independent data and identification strategies to show that the FLFPR decline is predominantly a demand-side phenomenon.

Deshpande and Singh (2021) use twelve rounds of a high-frequency household panel (the CMIE Consumer Pyramids survey) to decompose the fall in rural female LFPR. A Blinder–Oaxaca decomposition attributes almost none of the roughly ten-percentage-point decline over their study period to changes in supply-side characteristics such as education, household income, fertility, or other demographics. They further document that women's labour-market engagement is highly volatile, with women exiting and re-entering the labour force repeatedly over short horizons for reasons unrelated to marriage, childbirth, or income, and that women are disproportionately displaced from employment during negative shocks (for example, the 2016 demonetisation). The pattern is one of women being pushed out or unable to get in rather than voluntarily dropping out, which

is difficult to reconcile with a rising-income or conservative-norms account.

Deshpande and Singh (2024) make the demand-side case causal. Against the backdrop of India's structural transformation, in which the primary sector's share of GDP fell from about 24 percent in the late 1990s to roughly 17 percent by 2022 while services rose toward half of GDP, they instrument exogenous changes in local labour demand with a Bartik shift-share instrument and estimate the response of employment to demand at the district level. Female employment is highly responsive to local labour demand whereas male employment is not, and the period of declining FLFPR coincides with "jobless growth" in which women bore the brunt of stagnant employment creation. Their back-of-the-envelope accounting attributes the bulk of the post-2011 decline to weak local labour demand rather than to supply-side change, and they conclude that boosting FLFPR requires the creation of rural non-farm jobs accessible to women.

8.3 What the Caste-Norm Estimates Add

The IV estimate of the effect of caste norms on female labour force participation is statistically indistinguishable from zero (Table 16). This null bears on both explanations of the FLFP puzzle: a cultural backlash, in which gender norms worsened in ways not captured by attitudinal surveys, and the demand-side account above.

The cultural-backlash explanation finds little support in our data, with the caveat that a backlash invisible to attitudinal surveys cannot be ruled out with survey data. The norm indicators in Section 4 improve consistently across the period of declining participation, and we do not detect that weaker caste norms change participation.

The demand-side explanation is where the null is most informative, and reading it correctly requires keeping in mind that observed participation is an equilibrium of labour supply and labour demand. A zero causal effect of caste norms on participation therefore does *not* establish that caste norms leave women's labour supply untouched. It is equally consistent with a world in which caste norms do shift labour supply, through the education and marriage margins on which we find effects, but in which labour demand rather than norms determines whether that supply is absorbed into measured employment. When the jobs that historically employed rural women, overwhelmingly in agriculture, contract faster than accessible jobs are created elsewhere, a supply-side improvement need not show up as higher participation.

We therefore read the null as consistent with the structural-transformation account of Deshpande and Singh (2021, 2024) summarised above. Caste norms shape the human capital and marriage-market position with which women enter the labour market; whether that position translates into participation depends on a demand side that was contracting over the same period and that norms do not control. The evidence suggests that norms,

of either variety, are not the proximate driver of the FLFP decline.

8.4 Norm-Based Explanations Examined

Table 22: FLFP and gender norms, state and district regressions

	State panel, NFHS-2 to NFHS-5			District, NFHS-5	
	(1) Norm only	(2) + SNDP pc	SD of norm	(3) Norm only	SD of norm
Female education (years)	-0.016	-0.024	2.05	-0.010	1.68
	(0.020)	(0.028)		(0.010)	
Age at first birth	-0.005	-0.006	1.31	0.002	1.16
	(0.021)	(0.028)		(0.010)	
Age at marriage	0.001	0.002	1.57	0.008	1.46
	(0.020)	(0.025)		(0.011)	
Health agency	-0.013	-0.013	0.14	0.001	0.09
	(0.019)	(0.019)		(0.007)	
Mobility agency	0.040	0.042	0.23	0.006	0.09
	(0.026)	(0.027)		(0.006)	
Spousal education gap	0.000	0.000	1.07	-0.006	1.14
	(0.017)	(0.018)		(0.012)	
Spousal age gap	0.004	0.004	1.21	-0.015	1.39
	(0.019)	(0.019)		(0.014)	
Son preference	-0.012	-0.015	0.15	-0.012	0.10
	(0.019)	(0.021)		(0.008)	
DV justified	0.062***	0.065***	0.20	0.024*	0.20
	(0.011)	(0.011)		(0.012)	
Mean FLFP	0.301	0.301		0.268	
Observations	120	120		701	
Clusters (states)	33	33		33	
Round FE	Yes	Yes		–	
Region FE	No	No		Yes	
SNDP per capita	No	Yes		No	

Notes: Each cell is a separate OLS regression of FLFP (share of women aged 15–49 currently working) on one gender-norm indicator. Coefficients are the effect of a one-standard-deviation change in the indicator; the SD columns give that standard deviation in original units for each sample. Columns (1)–(2): state-by-round panel, NFHS-2, -3, -4 and -5, restricted to state-rounds with non-missing SNDP; column (2) adds RBI SNDP per capita (chained to 2011 prices) for the financial year before the survey, also scaled to one SD (Rs 57,217). With round FE alone, the SNDP coefficient is -0.001 (0.014). Column (3): district cross-section, NFHS-5, with region fixed effects (North omitted). Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A direct test of the norm-based explanation for low FLFPR is provided by Table 22, which regresses FLFPR on gender norm indicators at the state and district level. Each cell is a separate regression, with the coefficient scaled to a one-standard-deviation change in the indicator. In the state panel, which includes round fixed effects, eight of the nine indicators have no statistically significant association with FLFPR; the same holds across districts within regions. Without round fixed effects, the state coefficients on female education, health agency and mobility agency are negative and statistically significant, but that pattern reflects the common national trend (norm indicators improved while

Table 23: Correlation of non-veg diet (caste norms) with FLFP (EUS) at district-level - 2011

	(1) FLFP	(2) FLFP	(3) FLFP
% eat (fish)	-0.12***		
	(0.03)		
% eat (fish+meat)		0.02	
		(0.03)	
% UC eat (fish+meat)			-0.01
			(0.03)
Observations	586	586	571

Notes: Sample: Hindu households, district-level. Source: EUS 2011-12 (FLFP); HCES 2011-12 (diet proxy). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

participation fell) rather than differences across states.[5] Adding SNDP per capita leaves the norm coefficients almost unchanged, and SNDP itself is unrelated to FLFPR once round effects are absorbed (-0.001 per standard deviation, s.e. 0.014). We therefore find no sign in these data of the income channel behind the U-shaped female labour supply curve (Goldin, 1994).

The one exception is the share of women who justify domestic violence. A one-standard-deviation higher share is associated with FLFPR higher by 6.2 percentage points in the state panel ($p < 0.01$), about a fifth of the mean of 0.30, and by 2.4 percentage points across districts ($p < 0.1$). The sign runs against a norm-based account, in which more restrictive attitudes would accompany less work. One possible reading, which we do not test, is compositional: women in poorer and lower-caste households may be both more likely to work and more likely to report these attitudes. Taken together, the table suggests that differences in gender norms do not account for differences in FLFPR across states or districts.

Table 23 looks at the same question from the caste-norms side, across all districts. The fish-only share is negatively associated with FLFP (-0.12, $p < 0.01$), while the fish-and-meat and upper-caste measures show no statistically significant association (0.02 and -0.01). The fish-only association may reflect the eastern fish-eating belt, which this all-India sample includes and where the diet proxy does not measure caste orthodoxy. The two broader measures are consistent with the IV null.

9 Conclusion

This paper questions the one-dimensional narrative that attributes India's gender gaps to a static, monolithic patriarchal culture, and we close with the three contributions set

[5] Female education, for example, moves from -0.018 (s.e. 0.006) per year of schooling without round fixed effects to -0.008 (0.010) with them. Adding NFHS-3 to the panel or clustering standard errors by state, on its own, barely changes the estimate.

out in the introduction.

First, Indian gender norms have shifted substantially over two decades. Between 1998–99 and 2019–21, among ever-married women, female education rose by 3.2 years, age at first marriage and age at first birth by 1.8 years each, and stated justification of domestic violence fell by 12 percentage points. The pace of improvement differs across regions and caste groups, and caste gradients in gender outcomes are large and persistent.

Second, caste norms are associated with gender outcomes, and instrumental-variable estimates using the 1931 Brahman and Jain population shares suggest that the association is causal within the caste-purity belt. An interquartile weakening of caste dietary orthodoxy narrows the spousal education gap by 1.85 years and raises female education by 1.57 years, and weaker caste norms reduce son preference. These estimates control for the district's Scheduled Tribe population share, and the Hansen J test does not reject for any outcome. The education gap narrows because wives' education rises, which is consistent with the relaxation of caste endogamy but does not isolate it, and the spousal age gap widens because men's age at marriage rises by more than women's, consistent with a shift from kin-arranged marriage to marriage that rests on the groom's own earning capacity. Caste orthodoxy varies persistently across districts, and this variation predicts gender outcomes nine decades after the 1931 census.

Third, we do not detect a causal effect of caste norms on female labour force participation. Because participation is an equilibrium of labour supply and demand, this null is consistent with caste norms shaping supply through education and marriage while labour demand determines whether that supply is absorbed into measured work. Combined with the structural-transformation evidence in Deshpande and Singh (2021, 2024), this suggests that the FLFP decline tracks the contraction of agricultural employment more closely than changes in gender or caste attitudes.

Two policy implications follow, both tentative because we do not evaluate interventions. First, gender equality and the weakening of hierarchical social structures appear to be complements: gender-targeted interventions that leave the caste hierarchy intact may under-deliver where that hierarchy bounds the gender outcomes they can produce. Second, if caste norms are sticky—and a century of constitutional prohibition on untouchability, affirmative action, and economic growth has not dislodged them—then the policy lever may be to target caste norms directly rather than to wait for them to dissolve as a byproduct of other changes. Our findings suggest that the literature on gender in stratified societies would benefit from giving the second hierarchy as much attention as the first.

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A Appendix Figures

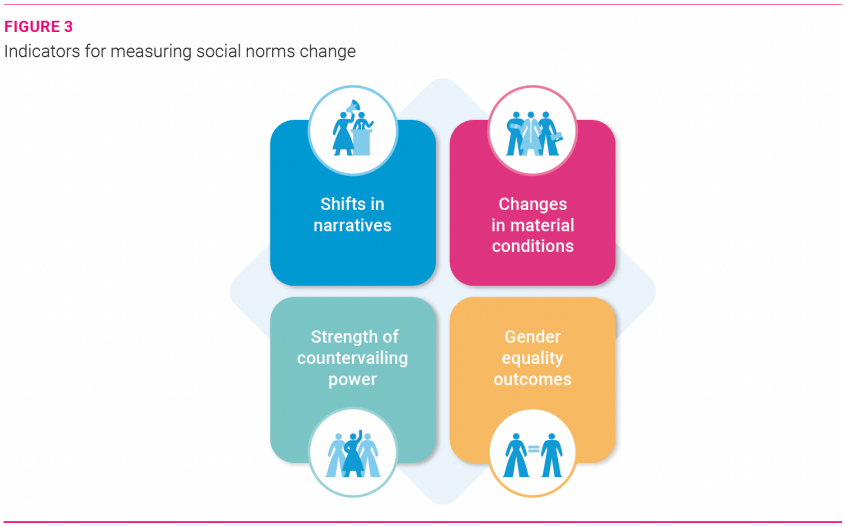


Figure A1: Indicators for measuring social norms change (UN Women, 2026, Figure 3). The framework distinguishes four sets of indicators: shifts in narratives, changes in material conditions, the strength of countervailing power, and gender equality outcomes.

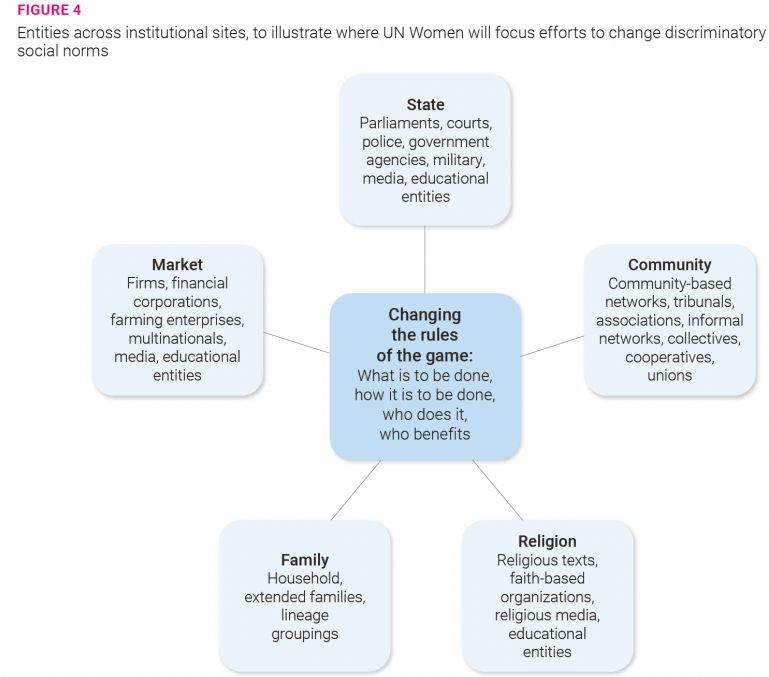


Figure A2: Institutional sites across which social norms are produced and enforced (UN Women, 2026, Figure 4): the state, market, community, family, and religion. Norms become "social" through these interlocking institutions, which is central to our argument that caste norms are sticky because they are institutionally embedded rather than merely attitudinal.

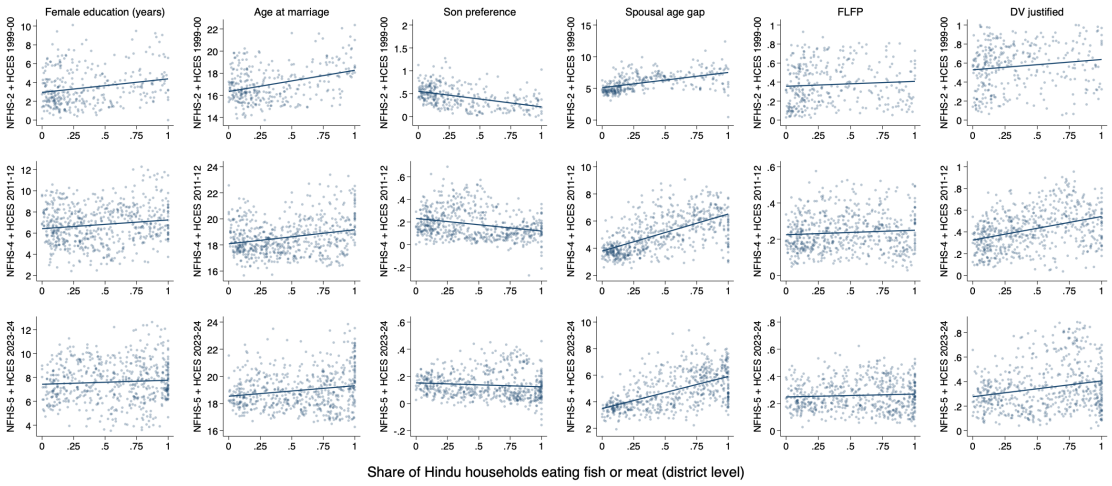


Figure A3: Co-evolution of diet (share of upper-caste Hindu households eating fish and meat) and gender norms at the district level, by round. Rows are survey rounds (1999, 2013, 2021); columns are gender outcomes. Each cell is a district-level scatter with a fitted line. The cross-sectional diet–gender relationship is present and stable within each round, the district-level counterpart to the national trajectories in Figures 2 and 3.

B Reconciling the Specifications: Sample versus Instrument

The three specifications vary the instrument and the sample at once, which on its own makes their differences hard to read. We therefore add the diagnostic cell that holds the sample fixed: the Brahman-only 2SLS estimated on the LIML intersection sample. Comparing this column with the Brahman-only full-sample column isolates the role of the *sample*; comparing it with the Jain-only and LIML columns, all on nearly the same sample, isolates the role of the *instrument*.

Table A1 reports this comparison for the two outcomes on which the design is most informative, female education and the spousal education gap, for the fish-and-meat (Panel B) and upper-caste (Panel C) proxies. Column (1) adds OLS on the intersection sample.

For the spousal education gap, the columns agree. With the fish-and-meat proxy, the Brahman-only estimate is -4.03 on the full sample and -3.97 on the intersection sample, close to the Jain-only (-4.40) and LIML (-4.06) estimates and larger in magnitude than OLS on the same sample (-2.43). For female education, the Brahman-only estimate changes sign between the full sample (-1.09) and the intersection sample ($+1.41$), but neither is statistically significant, and on the common sample its 95% confidence interval overlaps with the LIML estimate (3.44). The difference between the Brahman-only and LIML education estimates therefore reflects the sample and the weak Brahman first stage ($F = 8.4$ on the intersection sample) rather than a conflict between instruments. Panel C gives the same pattern. Read together with the identification-region argument of Section 6, this supports reading the three specifications as one finding, identified within the caste-purity belt.

Table A1: Reconciling the IV Specifications: Sample versus Instrument

	(1) OLS	(2) 2SLS	(3) 2SLS	(4) 2SLS	(5) LIML
Instrument(s)	None	Brahman	Brahman	Jain	Brahman + Jain
Sample	LIML	Full Brahman	LIML	Full Jain	LIML
Panel B: Share of Hindu HHs eating fish or meat					
Female education	1.449	-1.089	1.411	5.265***	3.438**
	(0.831)	(1.896)	(2.254)	(1.287)	(1.415)
First-stage F-stat	–	9.94	8.41	17.50	19.59
Hansen J p-value	–	–	–	–	0.164
Observations	294	521	294	314	294
Spouse education gap	-2.433***	-4.032***	-3.965***	-4.402***	-4.059***
	(0.590)	(0.666)	(0.940)	(0.629)	(0.638)
First-stage F-stat	–	9.94	8.41	17.50	19.59
Hansen J p-value	–	–	–	–	0.813
Observations	294	521	294	314	294
Panel C: Share of upper-caste Hindu HHs eating fish or meat					
Female education	1.197	-1.348	1.353	5.750***	3.834**
	(0.692)	(2.056)	(2.519)	(1.467)	(1.637)
First-stage F-stat	–	9.24	7.42	22.46	24.78
Hansen J p-value	–	–	–	–	0.145
Observations	288	514	288	308	288
Spouse education gap	-1.807***	-4.042***	-4.293***	-4.788***	-4.398***
	(0.496)	(0.731)	(1.040)	(0.773)	(0.771)
First-stage F-stat	–	9.24	7.42	22.46	24.78
Hansen J p-value	–	–	–	–	0.785
Observations	288	514	288	308	288

Notes: Coefficient on the diet proxy, standard errors clustered by state in parentheses. All columns use the same controls: district population, agricultural employment share, urban share, ST share and log night light density 2012. LIML sample is the estimation sample of column (5) for that outcome and proxy (districts with non-missing 1931 Brahman and Jain shares). Columns (2), (4) and (5) reproduce Tables A2, A4 and 16. Column (3) re-estimates column (2) on the LIML sample, so the difference between (2) and (3) reflects the sample and the difference between (3) and (5) reflects the added Jain instrument. First-stage F-stat: excluded-instrument F from the first stage on the full instrument sample in columns (2), (4) and (5) (joint test in column 5), and on the LIML sample in column (3). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Single-Instrument 2SLS Estimates

This appendix reports the single-instrument 2SLS estimates that serve as building blocks for the preferred LIML specification in Section 7.2.

Brahman-Only 2SLS

Table A2: IV Estimates: Effect of Caste Norms on Gender Norms (2SLS)

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	-1.483	0.195	-1.035	-0.017	0.626	-0.148	5.854***	-5.492***	0.036	-0.069
	(2.445)	(1.455)	(0.915)	(0.403)	(0.363)	(0.212)	(2.003)	(1.619)	(0.144)	(0.133)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	8.16	8.16	8.16	8.16	8.16	8.16	8.16	8.16	8.16	8.16
Observations	521	521	521	511	521	521	521	521	521	521
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	-1.089	0.143	-0.760	-0.012	0.459*	-0.108	4.298***	-4.032***	0.026	-0.051
	(1.896)	(1.054)	(0.712)	(0.293)	(0.238)	(0.142)	(0.748)	(0.666)	(0.104)	(0.100)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94	9.94
Observations	521	521	521	511	521	521	521	521	521	521
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	-1.348	0.103	-0.821	-0.034	0.478*	-0.101	4.336***	-4.042***	0.018	-0.062
	(2.056)	(1.133)	(0.780)	(0.315)	(0.255)	(0.153)	(0.861)	(0.731)	(0.113)	(0.109)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	9.24	9.24	9.24	9.24	9.24	9.24	9.24	9.24	9.24	9.24
Observations	514	514	514	504	514	514	514	514	514	514

Sample: Districts with non-missing 1931 Brahman share. Gender norms from NFHS-4 (2015-16), collapsed to district level, matched with caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). Instrument is 1931 Brahman population share, taken from Castelló-Climent et al. (2017). Controls included are district population (Census 2011), agricultural employment share, urban share, and scheduled tribe population share in 2011, and log night light density (SHRUG VIIRS, 2012) in 2012. FLFP is from PLFS 2011-12. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

Table A3: IV Estimates: Effect of Caste Norms on Gender Norms (2SLS)

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	-2.006	-0.461	-1.451	-0.153	0.626	-0.072	5.805***	-4.534***	0.010	-0.053
	(2.273)	(1.294)	(0.923)	(0.297)	(0.384)	(0.143)	(1.777)	(1.253)	(0.111)	(0.108)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	10.13	10.13	10.13	10.13	10.13	10.13	10.13	10.13	10.13	10.13
Observations	507	507	507	493	507	507	507	507	507	507
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	-1.469	-0.338	-1.063	-0.113	0.458	-0.052	4.250***	-3.320***	0.008	-0.039
	(1.833)	(0.982)	(0.777)	(0.233)	(0.275)	(0.101)	(0.807)	(0.665)	(0.081)	(0.081)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	8.30	8.30	8.30	8.30	8.30	8.30	8.30	8.30	8.30	8.30
Observations	507	507	507	493	507	507	507	507	507	507
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	-1.601	-0.498	-1.220	-0.141	0.444	-0.056	4.228***	-3.363***	0.010	-0.035
	(1.959)	(0.957)	(0.767)	(0.244)	(0.309)	(0.099)	(0.859)	(0.836)	(0.082)	(0.085)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	5.64	5.64	5.64	5.64	5.64	5.64	5.64	5.64	5.64	5.64
Observations	489	489	489	476	489	489	489	489	489	489

Sample: Districts with non-missing 1931 Brahman share. Gender norms from NFHS-5 (2019-21), collapsed to district level, matched with caste norms (diet proxy) from HCES 2023-24. Instrument is 1931 Brahman population share, taken from Castelló-Climent et al. (2017). Controls included are district population (Census 2011), agricultural employment share, urban share, and scheduled tribe population share in 2011, and log night light density (SHRUG VIIRS, 2012) in 2012. FLFP is from PLFS 2019. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

Tables A2 and A3 report 2SLS estimates using the 1931 Brahman share as the sole instrument, for NFHS-4 (2015–16) and NFHS-5 (2019–21) respectively.

Two results are consistent across the two survey rounds. First, weaker caste norms statistically significantly narrow the spousal education gap: for the fish-and-meat proxy, the estimated effect is -4.03 years in NFHS-4 and -3.32 years in NFHS-5 (both $p < 0.01$). Second, weaker caste norms are associated with a statistically significantly *larger* spousal age gap ($+4.30$ and $+4.25$ years, $p < 0.01$). The coefficient on female education is negative (-1.09 in NFHS-4 and -1.47 in NFHS-5) and not statistically significant. The coefficient on DV justification is positive and marginally statistically significant in NFHS-4 ($+0.46$, $p < 0.10$). The remaining coefficients, on marriage age, first birth age, FLFP, son preference, and female agency, are imprecisely estimated and not distinguishable from zero at conventional levels of statistical significance. Because the Brahman first stage is

weak (F below the Stock–Yogo threshold), 2SLS is biased toward OLS and its standard errors understate sampling uncertainty; we therefore do not read these coefficients as point estimates of the effect of caste norms. The first stage is correctly signed and statistically significant throughout the sample, so the imprecision is a matter of instrument strength rather than of a broken or wrong-signed first relationship. The estimates become more precise once we add the stronger Jain instrument, and Section B shows that, holding the sample fixed, the Brahman-only estimates are not statistically distinguishable from the Jain-only and LIML estimates.

Jain-Only 2SLS

Table A4: IV Estimates: Effect of Caste Norms on Gender Norms (2SLS, Jain 1931 IV)

	(1) Education	(2) Marriage Age	(3) First Birth Age	(4) FLFP	(5) DV Justified	(6) Son Preference	(7) Spouse Age Gap	(8) Spouse Educ Gap	(9) Mobility Agency	(10) Health Agency
Panel A: Share of Hindu HHs eating fish/seafood										
%eat(fish)	10.675***	3.406*	0.416	-0.185	1.184**	-0.222	8.792***	-8.926***	0.171	0.135
	(3.240)	(1.669)	(1.040)	(0.372)	(0.415)	(0.137)	(2.246)	(2.324)	(0.129)	(0.136)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	9.96	9.96	9.96	9.96	9.96	9.96	9.96	9.96	9.96	9.96
Observations	314	314	314	296	314	314	314	314	314	314
Panel B: Share of Hindu HHs eating fish or meat										
%eat(fish+meat)	5.265***	1.680*	0.205	-0.086	0.584***	-0.109*	4.336***	-4.402***	0.084	0.067
	(1.287)	(0.843)	(0.530)	(0.175)	(0.100)	(0.059)	(0.565)	(0.629)	(0.067)	(0.070)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50	17.50
Observations	314	314	314	296	314	314	314	314	314	314
Panel C: Share of Upper-Caste Hindu HHs eating fish or meat										
%UC_eat(fish+meat)	5.750***	1.849*	0.232	-0.111	0.617***	-0.121*	4.614***	-4.788***	0.089	0.071
	(1.467)	(0.971)	(0.597)	(0.192)	(0.124)	(0.068)	(0.561)	(0.773)	(0.078)	(0.080)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	22.46	22.46	22.46	22.46	22.46	22.46	22.46	22.46	22.46	22.46
Observations	308	308	308	290	308	308	308	308	308	308

Sample: Districts with non-missing 1931 Jain share (321 districts, 19 major states). Gender norms from NFHS-4 (2015-16), collapsed to district level, matched with caste norms (diet proxy) from HCES 2011-12 (NSS 68th round). Instrument is 1931 Jain population share (Census 1931), mapped to 2011 district boundaries via chained crosswalks (1931→1941→1951→2011). Controls included are district population (Census 2011), agricultural employment share, urban share, and scheduled tribe population share in 2011, and log night light density (SHRUG VIIRS, 2012) in 2012. FLFP is from PLFS 2011-12. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at state level in parentheses.

Table A4 reports 2SLS estimates using the 1931 Jain share as the sole instrument, restricted to the 19 major states for which the 1931 Jain share is observed. The first stage is considerably stronger than in the Brahman specification ($F = 17.5$ for the fish-and-meat proxy), and correspondingly more effects reach statistical significance.

First, weaker caste norms raise female education: a one-unit increase in the diet share (from no Hindu household to every Hindu household eating fish or meat) is associated with a +5.27-year increase in average female education ($p < 0.01$). Second, the spousal education gap narrows by 4.40 years ($p < 0.01$), and, consistent with the Brahman IV, the spousal age gap *widens* by +4.34 years ($p < 0.01$). Third, marriage age rises by +1.68 years ($p < 0.1$). Fourth, the coefficient on DV justification is positive and statistically significant (+0.58, $p < 0.01$), a pattern we interpret in Section 7.3. Effects on female labour force participation are small and not statistically different from zero.

D Appendix Tables

Cross-sectional correlations of diet and gender norms (full grid)

The following table reports the full cross-sectional grid of district-level correlations between each caste-norm diet measure and gender outcomes, across all states and both survey rounds. The drop-three-states version that we discuss in the main text appears in Section 5.1.

Table A5: Correlation of Hindu Diet Practices (Caste Norms Proxy) and Gender Norms at District-Year Level (NFHS 4 & 5; HCES 2011-12 and 2023-24)

	% eat (fish)		% eat (fish+meat)		% UC eat (fish+meat)	
	(1)	(2)	(3)	(4)	(5)	(6)
Female education (years)	0.69***	-0.22	0.56***	-0.36**	0.39***	-0.20*
	(0.17)	(0.16)	(0.16)	(0.14)	(0.15)	(0.11)
Age at marriage	1.16***	0.22*	0.91***	-0.13	0.69***	0.09
	(0.13)	(0.13)	(0.12)	(0.12)	(0.11)	(0.10)
FLFP	-0.12***	-0.13***	-0.02	-0.14***	-0.04**	-0.10***
	(0.02)	(0.05)	(0.02)	(0.05)	(0.02)	(0.04)
Mobility agency	0.07***	-0.02	0.08***	0.07***	0.06***	0.03
	(0.01)	(0.03)	(0.01)	(0.03)	(0.01)	(0.02)
Health agency	0.05***	-0.03	0.04***	0.05*	0.04***	0.02
	(0.01)	(0.03)	(0.01)	(0.02)	(0.01)	(0.02)
Son preference	-0.06***	-0.02	-0.07***	-0.05***	-0.06***	-0.04***
	(0.01)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)
Spousal age gap	2.20***	0.04	2.58***	0.15	2.23***	-0.00
	(0.12)	(0.19)	(0.10)	(0.17)	(0.10)	(0.14)
Spousal education gap	-1.81***	-0.04	-1.90***	-0.33**	-1.58***	-0.13
	(0.10)	(0.15)	(0.09)	(0.14)	(0.09)	(0.11)
DV justified	0.09***	-0.03	0.18***	0.00	0.13***	-0.01
	(0.02)	(0.04)	(0.02)	(0.03)	(0.02)	(0.03)
District FE	No	Yes	No	Yes	No	Yes
Observations	1199	1158	1199	1158	1152	1090

Notes: Each cell represents a separate regression of the row variable on the column's diet measure. UC refers to upper-caste only specification (non-SC/ST/OBC in the data). Diet measure is constructed as the proportion of Hindu households in a district with any positive expenditure on the given non-veg food item from the appended HCES rounds. Gender norms are at the district-year level, collapsed from appended NFHS rounds' women samples (aged 15–49). All gender norm definitions are same as earlier tables. FLFP round-matched (EUS 2011-12 for NFHS-4 districts, PLFS 2019 for NFHS-5 districts). Even-numbered columns include district fixed effects; odd-numbered columns include only NFHS round dummies. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. SE in parentheses.

NFHS question eligibility and harmonised samples

Table A6: NFHS-2 and NFHS-5: who answered each gender-norm question, and the harmonised samples

	NFHS-2 (1998–99)		NFHS-5 (2019–21)		
Indicator	Asked of	N	Asked of	N	Weight
Women's education (years)	All respondents	90,265	All women; ever-married used	542,830	v005
Age at first marriage	All respondents	90,303	Ever-married women	542,187	v005
Age at first birth	Women with a birth	80,869	Ever-married women with a birth	493,135	v005
Son preference	All respondents	85,501	All women; ever-married used	536,796	v005
Spousal education gap	All respondents (current or last husband)	90,045	State module, ever-married (current or last husband)	81,318	v005
Spousal age gap	Currently married	84,551	State module, currently married	76,910	v005
DV experienced	All respondents	90,294	DV module (one woman per household, privacy required)	63,851	v005 / d005
DV justified	All respondents	89,759	State module; ever-married used	81,211	v005
Health agency	All respondents	90,293	State module, currently married	76,910	v005
Mobility agency	All respondents	90,280	State module, currently married	76,910	v005

Notes: N is the number of ever-married women aged 15–49 with a non-missing value, the harmonised sample used in Section 4. NFHS-2 interviewed ever-married women only and asked every respondent the domestic violence questions. NFHS-5 interviewed all women; husband's education and age, decision-making and DV justification were asked only in the state-module subsample, and domestic violence only of the one woman per state-module household selected for the DV module, interviewed in private. Weights: women's sample weight (v005); for DV experienced in NFHS-5, the DV weight (d005).

Triple difference: region, caste and time

Table A7: Triple difference: education and reproductive timing norms by region, caste and time

	(1) Female education (years)	(2) Age at marriage	(3) Age at first birth	(4) Spousal education gap	(5) Spousal age gap
South/West States	-0.62	-0.35	-0.13	1.20***	-1.35***
	(0.77)	(0.47)	(0.40)	(0.38)	(0.38)
SC	-2.85***	-1.19***	-1.07***	-0.10	0.09
	(0.33)	(0.24)	(0.20)	(0.18)	(0.12)
ST	-3.58***	-1.20***	-1.16***	-0.17	-0.11
	(0.34)	(0.31)	(0.32)	(0.22)	(0.20)
OBC	-1.66***	-0.58	-0.45	0.11	-0.03
	(0.51)	(0.41)	(0.32)	(0.25)	(0.29)
South/West States × SC	-0.23	-0.30	-0.14	0.58**	-0.05
	(0.55)	(0.31)	(0.27)	(0.24)	(0.15)
South/West States × ST	1.26*	0.28	0.53	-0.17	0.06
	(0.65)	(0.59)	(0.43)	(0.36)	(0.31)
South/West States × OBC	-0.93	-0.75	-0.46	0.79**	-0.31
	(0.72)	(0.48)	(0.40)	(0.30)	(0.31)
Round 5	3.29***	1.80***	1.94***	-1.21***	-1.54**
	(0.49)	(0.46)	(0.44)	(0.39)	(0.61)
South/West States × Round 5	1.32	0.52	0.19	-1.60***	2.06***
	(0.96)	(0.74)	(0.69)	(0.52)	(0.73)
SC × Round 5	-0.36	-0.09	-0.02	0.54**	-0.31
	(0.54)	(0.27)	(0.23)	(0.20)	(0.18)
ST × Round 5	-0.41	0.32	0.35	0.75*	-0.46
	(0.59)	(0.49)	(0.49)	(0.38)	(0.37)
OBC × Round 5	-0.82	-0.51	-0.38	0.63**	-0.44
	(0.64)	(0.52)	(0.46)	(0.29)	(0.42)
South/West States × SC × Round 5	1.33	0.80*	0.36	-1.24***	0.38
	(0.86)	(0.44)	(0.41)	(0.41)	(0.36)
South/West States × ST × Round 5	-0.95	-0.68	-0.94	-0.09	-0.12
	(0.97)	(0.75)	(0.63)	(0.55)	(0.47)
South/West States × OBC × Round 5	2.37**	1.58**	0.96	-1.84***	1.26**
	(1.04)	(0.71)	(0.64)	(0.48)	(0.48)
Observations	603010	602449	546726	166425	156776

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). Spousal gaps use currently married women; in NFHS-5, husband's education and age come from the state-module subsample. OLS with full interactions of region (South/West vs North), caste group (General caste omitted) and survey round (NFHS-2 omitted); no individual controls. All variables are reported in years. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Triple difference: son preference, domestic violence and agency norms by region, caste and time

	(1) Son preference	(2) DV experienced	(3) DV justified	(4) Health agency	(5) Mobility agency
South/West States	0.28***	0.03*	-0.05	-0.03	-0.08*
	(0.07)	(0.02)	(0.08)	(0.04)	(0.05)
SC	0.10**	0.12***	0.08**	-0.03	0.01
	(0.04)	(0.03)	(0.04)	(0.02)	(0.04)
ST	0.13***	0.08***	0.13***	-0.04	0.02
	(0.03)	(0.02)	(0.04)	(0.04)	(0.03)
OBC	0.04	0.08***	0.13**	-0.02	0.05
	(0.07)	(0.02)	(0.06)	(0.03)	(0.06)
South/West States × SC	0.08*	-0.03	0.03	-0.06***	-0.03
	(0.04)	(0.03)	(0.04)	(0.02)	(0.04)
South/West States × ST	0.06	-0.03	0.01	0.01	0.04
	(0.06)	(0.03)	(0.04)	(0.06)	(0.09)
South/West States × OBC	0.12	-0.04*	-0.03	-0.04	-0.09
	(0.07)	(0.02)	(0.07)	(0.03)	(0.06)
Round 5	-0.15***	0.10***	-0.19**	0.29***	0.58***
	(0.05)	(0.02)	(0.07)	(0.03)	(0.04)
South/West States × Round 5	-0.36***	-0.08*	0.24*	0.03	0.10*
	(0.07)	(0.04)	(0.13)	(0.05)	(0.05)
SC × Round 5	-0.03	-0.01	-0.02	0.02	-0.02
	(0.05)	(0.04)	(0.04)	(0.02)	(0.04)
ST × Round 5	-0.07*	-0.02	-0.08*	0.04	-0.02
	(0.03)	(0.03)	(0.04)	(0.05)	(0.04)
OBC × Round 5	0.06	0.01	-0.09	0.01	-0.06
	(0.07)	(0.03)	(0.07)	(0.03)	(0.05)
South/West States × SC × Round 5	-0.14**	0.07	0.08	0.02	0.00
	(0.05)	(0.05)	(0.07)	(0.03)	(0.05)
South/West States × ST × Round 5	-0.05	0.12***	-0.02	-0.03	-0.09
	(0.07)	(0.04)	(0.08)	(0.07)	(0.10)
South/West States × OBC × Round 5	-0.20**	0.04	0.14	0.01	0.07
	(0.08)	(0.04)	(0.09)	(0.04)	(0.06)
Observations	592726	150043	166040	162466	162454

Notes: Sample is ever-married women aged 15–49 from NFHS rounds 2 and 5 (NFHS-2 interviewed only ever-married women), weighted by the women's sample weight (v005). DV experienced is the share ever physically hurt by the husband or in-laws, estimated on DV-module respondents with the DV weight (d005); sexual violence is excluded because NFHS-2 did not ask about it. DV justified uses the three scenarios asked in both rounds. In NFHS-5, the agency and DV-justification questions come from the state-module subsample. OLS with full interactions of region (South/West vs North), caste group (General caste omitted) and survey round (NFHS-2 omitted); no individual controls. Son preference is the difference between reported ideal number of boys and girls. All other outcomes are binary indicators of reported experience or attitudes. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interquartile effects and outcome means

Table A9: Implied Interquartile Effects of Caste Norms on Gender Norms (LIML)

	Panel A: Fish			Panel B: Fish or meat			Panel C: UC fish or meat		
Outcome	Coef.	IQR	Coef. × IQR	Coef.	IQR	Coef. × IQR	Coef.	IQR	Coef. × IQR
Education	8.230**	0.101	0.833	3.438**	0.456	1.569	3.834**	0.414	1.588
Marriage age	2.923	0.101	0.296	1.339	0.456	0.611	1.468	0.414	0.608
First birth age	0.120	0.101	0.012	0.051	0.456	0.023	0.062	0.414	0.026
FLFP	0.503	0.101	0.051	0.178	0.456	0.081	0.185	0.414	0.077
DV justified	1.393***	0.101	0.141	0.634***	0.456	0.289	0.669***	0.414	0.277
Son preference	-0.308	0.101	-0.031	-0.142*	0.456	-0.065	-0.155*	0.414	-0.064
Spouse age gap	9.964***	0.101	1.008	4.572***	0.456	2.086	4.879***	0.414	2.021
Spouse educ gap	-8.803***	0.101	-0.891	-4.059***	0.456	-1.852	-4.398***	0.414	-1.822
Mobility agency	0.150	0.101	0.015	0.069	0.456	0.032	0.071	0.414	0.029
Health agency	0.013	0.101	0.001	0.004	0.456	0.002	-0.001	0.414	-0.000
Diet proxy p25 / p75	0.010 / 0.111			0.105 / 0.562			0.071 / 0.486		
Districts (education sample)	294			294			288		

Notes: Coef. is the LIML coefficient on the diet proxy from Table 16 (instruments: 1931 Brahman and Jain population shares; controls: district population, agricultural employment share, urban share, ST share, log night light density 2012; standard errors clustered by state). IQR is the interquartile range (p75 minus p25) of the diet proxy in the LIML estimation sample for female education in that panel. Coef. × IQR is the implied change in the outcome from moving a district from the 25th to the 75th percentile of the diet proxy. Diet proxies are shares (0 to 1) of Hindu households (Panel C: upper-caste Hindu households) reporting any fish (Panel A) or any fish or meat (Panels B and C) in the last 7 days, HCES 2011–12. Stars refer to the coefficient. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Gender Norm Outcomes in the LIML Estimation Sample

Outcome	Mean	SD	Districts
Education	7.026	1.818	294
Marriage age	18.497	1.243	294
First birth age	20.470	0.953	294
FLFP	0.454	0.212	287
DV justified	0.427	0.196	294
Son preference	0.134	0.089	294
Spouse age gap	4.985	1.472	294
Spouse educ gap	1.535	1.191	294
Mobility agency	0.751	0.090	294
Health agency	0.741	0.092	294

Notes: District-level means and standard deviations (unweighted across districts) in the Panel B LIML estimation sample of Table 16 for each outcome. Outcomes are district means from NFHS-4 (2015–16) except FLFP (EUS 2011–12). Education and spouse education gap in years; marriage age, first birth age and spouse age gap in years; son preference is ideal number of boys minus ideal number of girls; DV justified, mobility agency, health agency and FLFP are shares.

E Do Economic Factors Explain Change in Norms?

A word on the unit of analysis is in order, because this section departs from the rest of the paper. The two economic correlates we examine here, per-capita income and social-sector expenditure, are budgetary and national-accounts aggregates that are compiled and reported by governments at the state level; neither is available as a consistent district-level series over our study period. We therefore conduct this analysis at the state level. Everywhere else, where the underlying data permit it, we work at the district level, which is the level at which our caste-norm proxies and our historical instruments vary and the

level at which the central results of the paper are estimated. The state-level results in this section should accordingly be read as broad descriptive associations rather than as estimates comparable in resolution to the district-level analysis in the main text.

E.1 State Income and Gender Norms

We regress gender norm outcomes on state net domestic product per capita (Reserve Bank of India, various years), mapping each NFHS round to the contemporaneous (and one-year lagged) SNDP figure:

$$y_{st} = \beta_0 + \beta_1 \, \text{SNDP}_{st} + \varepsilon_{st} \tag{E.1}$$

Table A11: Impact of SNDP per capita and 1-year Lagged SNDP Per Capita on Gender Norms

	State Net Domestic Product	
Gender Norms	**SNDPC**	**SNDPC Lagged**
Female Education	0.0000276***	0.0000259***
	(0.00)	(0.00)
Marriage Age	0.0000157***	0.0000152***
	(0.00)	(0.00)
First Birth Age	0.0000151***	0.0000143***
	(0.00)	(0.00)
Education Gap	-8.47e-06***	-8.87e-06***
	(0.00)	(0.00)
Age Gap	-6.21e-06***	-5.07e-06***
	(0.00)	(0.00)
Son Preference	-1.77e-06***	-1.44e-06***
	(0.00)	(0.00)
Domestic Violence (DV)	-7.50e-07***	-8.02e-07***
	(0.00)	(0.00)
DV Justification	-1.36e-06***	-1.45e-06***
	(0.00)	(0.00)
Health Agency	1.05e-06***	1.13e-06***
	(0.00)	(0.00)
Mobility Agency	1.86e-06***	1.98e-06***
	(0.00)	(0.00)

Note: Each cell gives the coefficient from running the regression of the gender norm in the row on the contemporaneous (SNDPC) or lagged State Net Domestic Product per capita (SNDPC Lagged). Sample for gender norms is women aged 15–49 from NFHS rounds 1, 2, 3, 4, and 5—collapsed at the state-year level. SNDPC for corresponding years is taken from RBI's *Handbook of Statistics on Indian States*.
*** p<0.01, ** p<0.05, * p<0.1. SE reported in parentheses.

Table A11 reports a consistently positive association between state income and gender outcomes: a Rs 1,000 increase in per-capita SNDP is associated with 0.026 additional

years of female education, 0.016 additional years of marriage age, and a 0.136 percentage point reduction in DV justification (all statistically significant at 1%). While these associations are informative, they suffer from obvious endogeneity: richer states may have better gender norms for reasons unrelated to income growth. Causal estimation of income effects on norms remains an important direction for future work.

E.2 Social Sector Expenditure and Norms

We ask whether government spending on social programmes moves the needle on gender norms, estimating:

$$y_{ist} = \beta_0 + \beta_1 \, \text{SSE}_{st} + \alpha_t + \varepsilon_{ist} \tag{E.2}$$

where SSE is the share of state total disbursement allocated to the social sector (Reserve Bank of India, various years) and α_t are round fixed effects.

Table A12: Gender norms regressed on Social Sector Expenditures - 1

	(1) fem_educ b/se	(2) fem_marriageage b/se	(3) fem_firstbirthage b/se	(4) spouse_educgap b/se	(5) spouse_agegap b/se
sse_budget	-0.08***	-0.09***	-0.07***	0.01	0.04*
	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
Observations	1424159	1046309	967961	250319	236995

Note: Sample for gender norms (measured in years) is women aged 15–49 from NFHS rounds 3, 4, and 5. Independent variable is the social sector expenditure of a state as percentage of its total disbursement in the year corresponding to a NFHS round and is taken from RBI's Handbook of Statistics on Indian States.
*** p<0.01, ** p<0.05, * p<0.1. SE reported in parentheses.

Table A13: Gender norms regressed on Social Sector Expenditures - 2

	(1) son_preference b/se	(2) domesticviolence b/se	(3) dv_justify b/se	(4) fem_healthagency b/se	(5) fem_mobilityagency b/se
sse_budget	0.0019	0.0008**	-0.0029	-0.0041***	-0.0027
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Observations	1404089	1424171	327650	236238	236236

Note: Sample for gender norms is women aged 15–49 from NFHS rounds 3, 4, and 5. Son preference is the difference between reported ideal number of boys and girls. All other outcomes are binary indicators of reported experience or attitudes. Independent variable is the social sector expenditure of a state as percentage of its total disbursement in the year corresponding to a NFHS round and is taken from RBI's Handbook of Statistics on Indian States.
*** p<0.01, ** p<0.05, * p<0.1. SE reported in parentheses.

The results in Tables A12 and A13 are initially puzzling: higher SSE shares are associated with worse gender outcomes (lower female education, lower marriage and birth ages, reduced agency). However, this pattern most likely reflects reverse causation rather than a perverse causal effect of spending. States with more regressive gender norms spend more on remedial social programmes. Bihar, for instance, invests heavily in schemes to keep girls in school because its gender outcomes are poor. This "policies follow problems"

pattern is a canonical omitted-variable challenge; disentangling it is part of our ongoing causal analysis.